



Controlling Phosphorus in Aquatic Systems: A Mini Review on Emerging Adsorption and Electro-Assisted Hybrid Technologies for Sustainable Water Treatment

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Abstract

Phosphorus management has become a major challenge in wastewater treatment because phosphorus is both a key driver of eutrophication and a finite nutrient resource that should be recovered and reused. This review examines recent advances in phosphorus removal and recovery, with particular emphasis on emerging adsorption-based materials, membrane processes, and electro-assisted hybrid technologies. Conventional approaches, including chemical precipitation and enhanced biological phosphorus removal, are first considered to establish their performance and limitations as a basis for comparison with emerging treatment pathways. Recent developments in adsorption are then critically evaluated, ranging from metal oxides and biochar-based materials to layered double hydroxides, metal-organic frameworks, and engineered composite adsorbents. Particular attention is given to the relationships among material design, phosphate-binding mechanisms, selectivity, regeneration, and performance under realistic wastewater conditions. Membrane and electro-assisted systems are further examined, including electrochemical-membrane configurations, capacitive deionization, and membrane capacitive deionization, which provide opportunities for electrically controlled separation, phosphorus enrichment, and regeneration. Rather than considering removal efficiency alone, the review highlights key trade-offs involving selectivity, stability, fouling, energy demand, scalability, and the ability to convert separated phosphorus into a recoverable form. Emerging hybrid architectures that integrate selective adsorption, membrane separation, and electrochemical regeneration, therefore, represent an important transition from conventional phosphorus removal toward multifunctional treatment platforms that bridge high-efficiency phosphorus polishing and circular-economy nutrient recovery.

Keywords Adsorption · Biosorption · Membrane capacitive deionization · Phosphorus removal · Phosphorus recovery · Wastewater treatment

Introduction

Water Pollution and Environmental Concerns

Securing access to clean, potable water for future generations remains a critical global challenge, particularly amid rapid population growth, increasing pollution of water resources, and rising water demand. The quality of natural water resources is strongly influenced by the efficiency and operational performance of wastewater treatment plants within a region. Wastewater generally includes sewage generated from domestic activities, as well as effluent discharged from industrial facilities and commercial

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complexes [1, 2]. Such wastewater streams often contain a wide range of contaminants, including organic matter, suspended solids, heavy metals, pharmaceuticals, and nutrient compounds such as nitrogen (N) and phosphorus (P). If these pollutants are not properly removed before discharge, they can significantly degrade aquatic ecosystems and threaten water quality [3]. It is essential to identify the nature of water pollutants and evaluate their impacts on both human health and the environment. Plastics [4], heavy metals [5], pharmaceutical residues [6], industrial dyes [7], as well as nitrogen and phosphate (PO_4^{3-}) compounds, are major water contaminants that seriously affect our lives. Different countries have established their own permissible concentration limits for these contaminants in surface waters and wastewater discharges according to national environmental regulations [8]. Among these contaminants, phosphorus has received particular attention due to its dual environmental and resource-related significance.

Phosphorus as a Pollutant and Resource

Phosphorus, an essential element for plant growth, can also act as a pollutant when its concentration exceeds regulatory limits. Elevated phosphorus levels have been associated with potential health risks, including cardiovascular and kidney-related disorders. In aquatic environments, phosphorus is predominantly present in the form of phosphate ions (PO_4^{3-}).

Typically, phosphorus concentrations in industrial effluents range from 4 to 10 mg/L [9]. However, when phosphate concentrations exceed established thresholds, often above 0.1 mg/L, nutrient accumulation occurs, leading to eutrophication in aquatic systems. This process promotes excessive algal growth, which, in turn, reduces dissolved oxygen levels and deteriorates overall water quality, thereby disrupting the ecological balance. In terrestrial systems, excessive soil phosphorus can also hinder the uptake of essential micro-nutrients such as iron and zinc, ultimately affecting agricultural productivity [10–13]. Therefore, the development of efficient strategies for controlling and removing phosphorus from aquatic environments has become a central focus in wastewater treatment research. Wastewater treatment generally consists of several sequential stages, including pretreatment, primary treatment, secondary treatment, and tertiary treatment, as illustrated in Fig. 1(a) [14]. In addition to its role as a pollutant, phosphorus is increasingly recognized as a valuable resource that can be recovered and reused, contributing to environmental sustainability and resource efficiency. The broader significance of nutrient recovery is illustrated in Fig. 2(a), highlighting its multifaceted benefits, including improvements in water quality, protection of aquatic ecosystems, enhancement of agricultural productivity through fertilizer reuse, conservation of finite phosphorus resources, and contributions to food security. Furthermore, nutrient recovery supports public health and

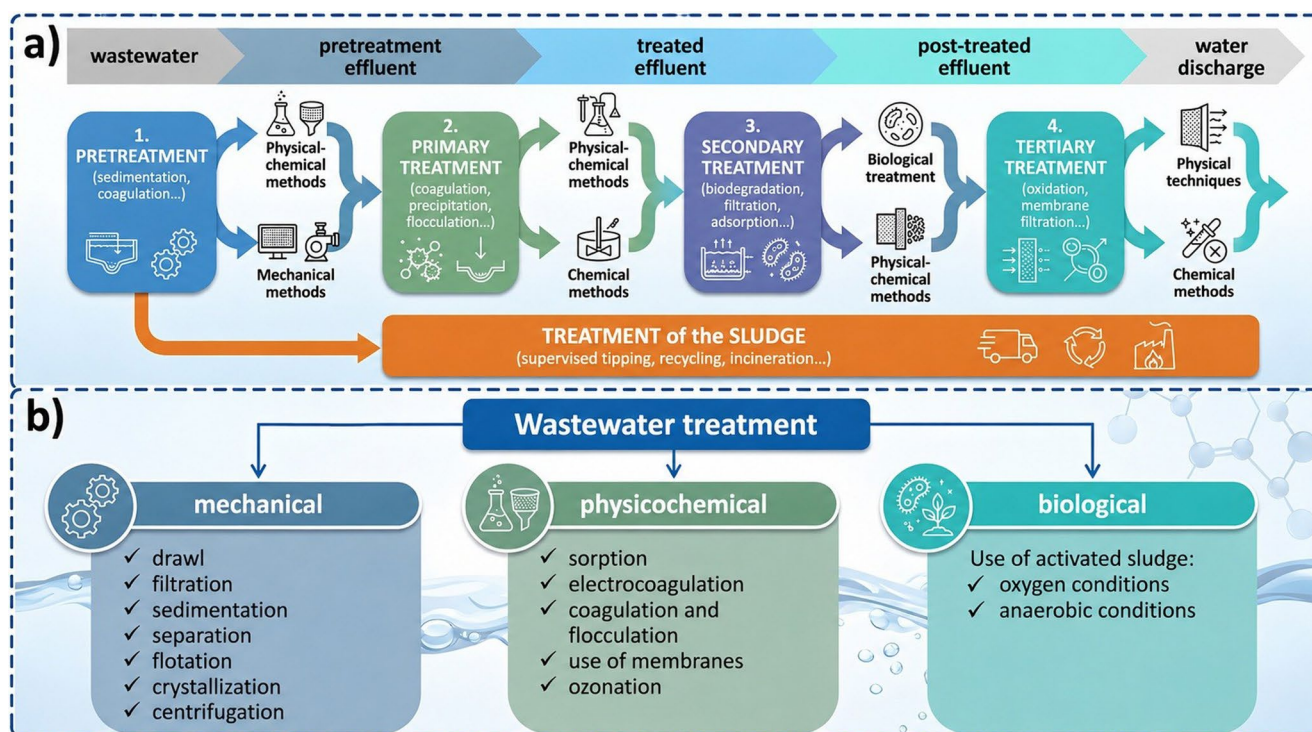


Fig. 1 (a) General stages of wastewater treatment processes showing the sequence of pretreatment, primary, secondary, and tertiary treatment stages [14], and (b) Main wastewater treatment technologies, including mechanical, physicochemical, and biological [14]

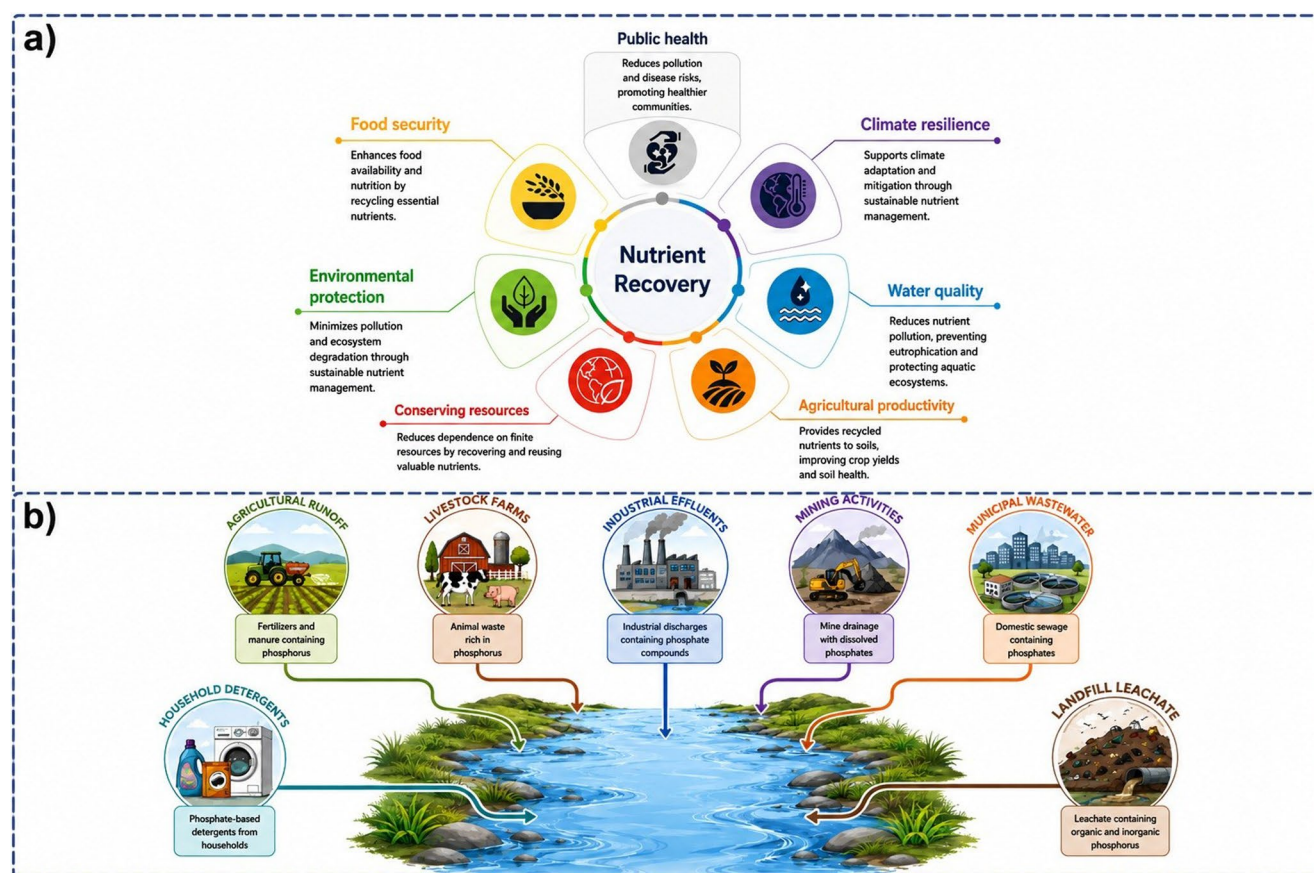


Fig. 2 (a) Environmental and societal benefits of nutrient recovery [15], and (b) Main sources of Phosphorus

strengthens climate resilience by reducing environmental pollution and promoting sustainable resource management practices [15]. This dual role underpins the increasing interest in sustainable phosphorus management strategies.

Sources and Regulatory Considerations

Correspondingly, it is critical to regularly monitor phosphorus concentrations in aquatic environments to maintain the natural balance of nutrient cycles within ecosystems. Phosphorus can leach into aquatic environments through the extensive use of phosphorus-based fertilizers in agricultural soils to improve crop yields. Similarly, phosphorus can be released into aquatic environments from both natural and anthropogenic sources, including animal waste, landfill leachate, industrial wastewater streams, soil erosion, urban runoff from fertilized agricultural lands, and household cleaning products, as illustrated in Fig. 2(b) [16–18]. Due to the ecological risks associated with phosphorus enrichment, regulatory authorities have established strict concentration limits for phosphorus in natural waters. According to the Australian water quality guidelines for fresh and marine waters, the recommended limits for phosphorus concentrations in different

aquatic environments are summarized in Table 1 in order to prevent excessive plant and algal growth [19]:

In addition to regional guidelines, several countries and regulatory bodies have established strict ambient phosphorus standards to control eutrophication in surface waters. For example, allowable phosphorus concentrations typically range from 0.02 to 0.4 mg/L in China, while limits of 0.05 to 0.1 mg/L are commonly applied in the United States, depending on the receiving water body. Similarly, stringent standards have been implemented in the European Union, New Zealand, Australia, and Taiwan, where allowable phosphorus concentrations often fall below 0.1 mg/L to protect aquatic ecosystems from nutrient enrichment [20, 21].

Table 1 Concentration limit for P in different aquatic environments

Aquatic Environment	Concentration range ($\mu\text{g/L}$)	Measurement basis
Rivers and streams	10–100 $\mu\text{g/L}$	as total phosphorus (TP)
Lakes and reservoirs	5–50 $\mu\text{g/L}$	as total phosphorus (TP)
Estuaries and embayments	5–15 $\mu\text{g/L}$	as phosphate-phosphorus ($\text{PO}_4\text{-P}$)
Coastal waters	1–10 $\mu\text{g/L}$	as phosphate-phosphorus ($\text{PO}_4\text{-P}$)

Recent reviews have provided valuable insights into different aspects of phosphorus removal and recovery. Some have focused primarily on adsorption materials, including biochar, metal oxides, layered double hydroxides (LDHs), and metal-organic frameworks (MOFs), whereas others have examined membrane-based separation or electro-assisted technologies such as electrosorption, electrocoagulation, CDI, and MCDI [22–27]. For example, Y. Ren et al. [27] reviewed electrochemical phosphorus removal and recovery through electrosorption, electrocoagulation, and electro-induced precipitation, while also drawing attention to less frequently considered phosphorus species, including non-orthophosphate and phosphite. However, despite these advances, the available literature remains largely divided among material-specific adsorption reviews, membrane-focused studies, and reviews of individual electrochemical processes. Consequently, comparatively little attention has been given to how material characteristics and process configurations collectively determine whether phosphorus is merely removed from water, concentrated in a secondary stream, released during regeneration, or ultimately recovered as a reusable product. The differences in scope and emphasis between these recent reviews and the present review are summarized in Table 2.

Against this background, the present review adopts an integrated material-to-process perspective. It follows the phosphorus-management pathway from adsorbent design and selective phosphate capture to membrane separation, electrochemical regeneration, phosphorus enrichment, and subsequent recovery. Particular attention is given to the relationships among material design, phosphate selectivity, regeneration, stability, energy demand, performance in real wastewater, and scalability. In contrast to reviews that treat removal efficiency as the principal indicator of treatment success, this review distinguishes phosphorus removal, concentration, regeneration, and recovery as separate process outcomes. By examining adsorption, membrane separation, and electro-assisted technologies within a common framework, the review further evaluates how hybrid architecture

can meet stringent phosphorus-polishing requirements while supporting circular nutrient management.

Literature Search and Review Methodology

A structured literature search was conducted to identify relevant studies on emerging technologies for phosphorus removal and recovery from water and wastewater. The literature search was performed using *Web of Science*, covering all publications from the start of 2003 to the end of 2025, with the final search conducted in August 2026. Search terms were constructed around three main themes: phosphorus removal and recovery, adsorption-based treatment, and electro-assisted/membrane-based technologies. Representative search terms included combinations of “phosphorus removal”, “phosphate removal”, “phosphorus recovery”, “phosphate recovery”, “adsorption”, “adsorbent”, “biochar”, “metal oxide”, “layered double hydroxide”, “metal-organic framework”, “membrane”, “electrochemical”, “electrocoagulation”, “capacitive deionization”, “CDI”, and “membrane capacitive deionization” or “MCDI”. Boolean operators (AND/OR) were used to combine related terms and refine the search. Studies were considered eligible when they (i) investigated phosphorus or phosphate removal, separation, concentration, or recovery from water or wastewater; (ii) addressed adsorption-based, membrane-based, electrochemical, or hybrid treatment technologies relevant to the scope of this review; and (iii) reported experimental performance, mechanistic information, material characteristics, regeneration behavior, recovery performance, or process-level considerations.

To complement the structured literature review and illustrate the broader development of research activity in this field, a separate bibliometric search was conducted using *Web of Science* in August 2026. The search was performed using the Topic query: (phosphorus OR phosphate OR orthophosphate OR “phosphate-P” OR nutrient) NEAR/5 (remov OR captur* OR recover* OR separ* OR extract* OR concentrat* OR adsor*) AND (water OR wastewater

Table 2 Scope and distinguishing perspectives of recent reviews on phosphorus removal and recovery compared with the present review

Recent review	Adsorbents	Membranes	Electro-assisted/CDI	Removal vs. recovery distinction	Regeneration/energy/real-wastewater evaluation	Integrated material-process perspective
[22]	—	Partial	✓	Limited	Partial	Partial
[23]	✓	—	—	Partial	Partial	—
[26]	✓	—	—	Partial	✓	—
[24]	✓	—	—	✓	✓	Partial
[27]	—	Partial	✓	✓	Partial	Partial
[28]	—	✓	✓	Partial	Partial	Partial
Present review	✓	✓	✓	✓	✓	✓

OR “waste water” OR effluent OR sewage). The search returned 146,347 records. Annual publication counts were obtained using the *Web of Science* “Analyze Results” function, and complete publication years from 2003 to 2025 were used to generate Fig. 3. The annual number of retrieved publications increased from 1,957 in 2003 to 10,168 in 2025, corresponding to more than a fivefold increase in annual research output. The strongest expansion occurred during the most recent decade, reflecting growing attention to nutrient-pollution control, phosphorus capture and recovery, and advanced separation technologies. Within this broader trend, research emphasis has shifted from bulk removal to selective adsorption, membrane separation, electro-assisted treatment, concentration, regeneration, and resource recovery, which are the central focus of the present review. The bibliometric analysis was conducted separately from the literature-screening procedure used to identify studies for detailed critical evaluation.

Overview of Phosphorus Removal Technologies

Overview of Conventional and Emerging Phosphorus Removal Technologies

Meeting increasingly stringent discharge limits requires the implementation of highly efficient phosphorus removal technologies [29]. Established phosphorus-removal approaches

include chemical precipitation and enhanced biological phosphorus removal (EBPR), both of which are widely applied in conventional wastewater treatment. In contrast, adsorption and membrane-based processes are increasingly used as advanced physicochemical treatment or polishing technologies, particularly where low residual phosphorus concentrations or subsequent resource recovery are targeted. Wastewater treatment technologies can generally be classified into mechanical, physicochemical, and biological processes, as illustrated in Fig. 1(b) [14].

These technologies differ considerably in their removal mechanisms, achievable effluent quality, operational requirements, and potential for phosphorus recovery. Chemical precipitation typically achieves phosphate concentrations in the treated wastewater of approximately 0.5–1 mg/L, while EBPR can reduce it to 0.1–0.2 mg/L under optimal conditions. Advanced polishing approaches, including adsorption and membrane filtration, have been reported to achieve substantially lower residual phosphate concentrations, in some cases below 0.01 mg/L under suitable operating conditions. Adsorption also allows phosphorus release during adsorbent regeneration, although successful removal does not guarantee subsequent phosphorus recovery [20, 30]. Nevertheless, each technology has inherent limitations in efficiency, cost, operational stability, and recovery potential, particularly under real wastewater conditions.

A general comparison of the major phosphorus removal technologies used in wastewater treatment is summarized in Table 3.



Fig. 3 Annual publication trend between 2003 and 2025 based on a *Web of Science* search as described in Sect. 1.4

Table 3 Comparison of common phosphorus removal technologies used in wastewater treatment

Method	Removal Mechanism	Typical Efficiency	Advantages	Limitations	Ref.
Chemical precipitation	Formation of insoluble phosphate salts using metal ions such as Fe^{3+} , Al^{3+} , or Ca^{2+}	80–99%	Simple operation and widely applied in municipal wastewater treatment	High sludge generation, chemical consumption, and limited phosphorus recovery	[31]
Biological treatment (EBPR)	Phosphorus uptake and storage by polyphosphate-accumulating organisms (PAOs)	70–90%	Environmentally friendly and reduced chemical usage	Sensitive to operational conditions and process instability	[32, 33]
Adsorption	Phosphate ions bind to the surface of adsorbent materials (e.g., metal oxides, activated carbon, biochar)	Up to 95%	Effective at low phosphorus concentrations; potential for phosphorus recovery	High treatment cost and adsorbent regeneration requirement	[21, 32]
Membrane separation	Physical separation via membrane processes such as nanofiltration or reverse osmosis	90–99%	High selectivity and capable of achieving very low effluent phosphorus levels	High energy demand and membrane fouling	[34]

Scope and Focus of this Review

The 1950s are generally regarded as the beginning of systematic efforts to develop technologies for the removal of phosphorus from wastewater. This shift was mainly driven by increasing concerns about eutrophication and the urgent need to control its concentrations in surface waters [35]. In the early stages of technological development, chemical precipitation emerged as the dominant and most practical method for phosphorus removal, and it remains one of the most effective and widely applied techniques in wastewater treatment today. Over the past several decades, however, advances in environmental engineering and material science have encouraged the development of alternative and more advanced treatment technologies aimed at improving removal efficiency, operational flexibility, and phosphorus recovery potential [35, 36].

Phosphorus-removal technologies can broadly be classified into adsorption, biological treatment, membrane separation, chemical precipitation, and electro-assisted processes, as illustrated in Fig. 4(a) [37–40]. While chemical precipitation and EBPR represent established treatment pathways, recent developments increasingly emphasize adsorption, membrane separation, and electro-assisted or hybrid systems for advanced phosphorus polishing and recovery. In this review, conventional chemical and biological processes are considered primarily as a reference framework for understanding the limitations that have motivated the development of advanced treatment technologies. The main focus is therefore placed on emerging adsorption-based materials and electro-assisted membrane and hybrid systems, particularly technologies that offer opportunities for selective phosphorus polishing, concentration, regeneration, and recovery.

Rather than providing an exhaustive review of all phosphorus-removal technologies, this review examines the progression from engineered adsorbents to integrated electro-assisted separation systems and critically considers how material design and process configuration influence selectivity, regeneration, energy demand, stability, scalability, and performance under realistic wastewater conditions. Particular attention is given to adsorption-based systems and electro-assisted technologies, including electrocoagulation, electrodialysis, CDI, MCDI, and related hybrid configurations [15, 18, 29, 41]. Importantly, the review evaluates these technologies not only for their ability to remove phosphorus from the aqueous phase but also for their capacity to enable subsequent phosphorus concentration, regeneration, and recovery.

In this review, phosphorus removal and phosphorus recovery are treated as related but distinct outcomes. Phosphorus removal refers to the transfer of phosphorus from the treated aqueous phase to another phase, such as an adsorbent, membrane concentrate, biomass, sludge, or precipitate. Removal alone, therefore, does not demonstrate resource recovery. Phosphorus concentration refers to enrichment of the separated phosphorus into a smaller-volume stream or phase, while regeneration or desorption refers to the release of phosphorus from a sorbent or electrode to restore treatment capacity. Phosphorus recovery is considered to occur only when the separated phosphorus is subsequently collected in a form or stream that can reasonably support reuse or conversion into a usable phosphorus-containing product. Accordingly, removal efficiency, regeneration efficiency, concentration factor, and recovery efficiency are treated as separate performance indicators throughout this review.

Throughout this review, phosphorus concentrations are reported using the terminology and analytical basis specified

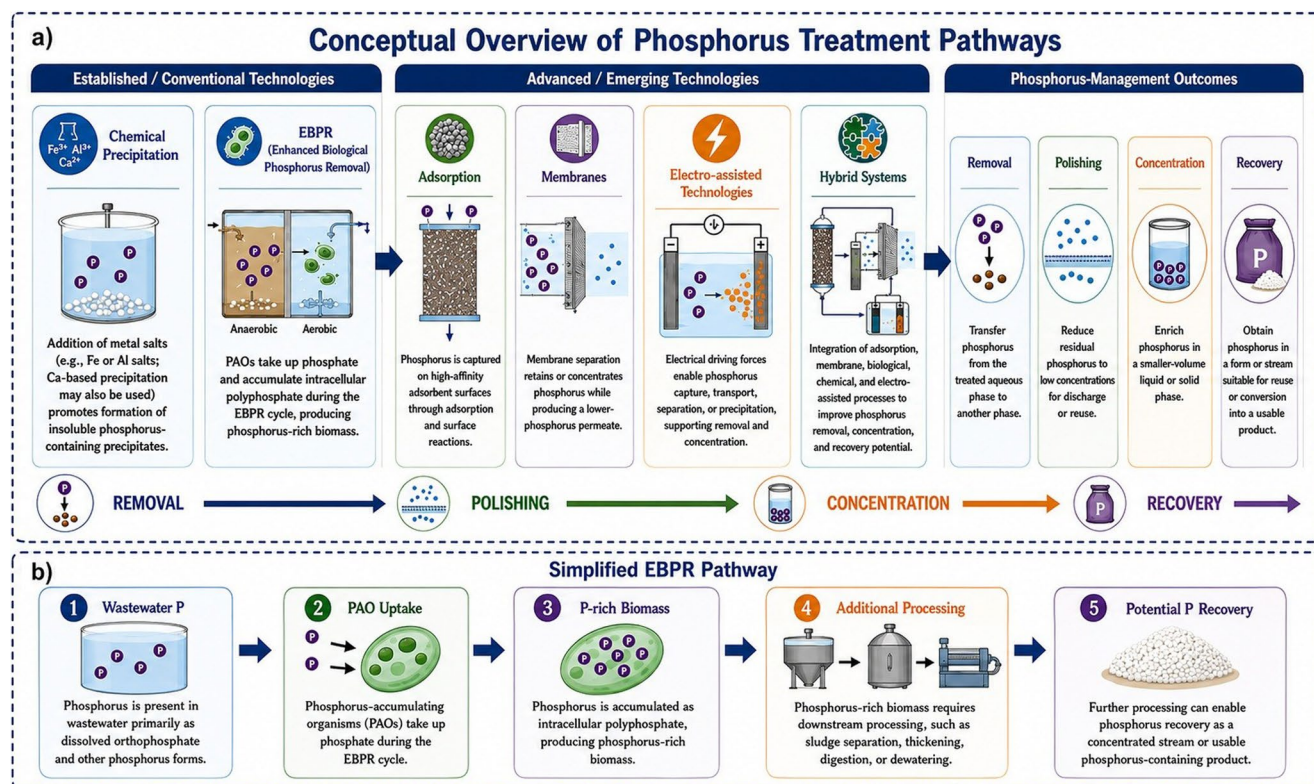


Fig. 4 Conceptual overview of phosphorus treatment pathways: (a) classification and progression from established phosphorus-removal methods toward advanced adsorption, membrane, electro-assisted,

and hybrid technologies; and (b) simplified EBPR pathway showing biological phosphorus uptake and accumulation in phosphorus-rich biomass prior to downstream recovery

in the original studies. Total phosphorus (TP), orthophosphate, phosphate expressed as phosphorus ($\text{PO}_4\text{-P}$), and phosphate expressed as PO_4^{3-} are therefore distinguished and are not treated as directly interchangeable concentration units. Where comparison among studies is made, the reported phosphorus basis is explicitly identified to avoid ambiguity arising from differences in analytical methods or concentration conventions. Accordingly, the technologies discussed in this review are critically evaluated beyond removal percentage alone. Where information is available, comparison considers phosphate selectivity in the presence of competing ions, regeneration and repeated-cycle performance, material or membrane stability, energy and chemical requirements, performance in real wastewater, scalability, and the form and concentration of phosphorus generated after treatment. These criteria are particularly important because high removal efficiency under controlled laboratory conditions does not necessarily indicate practical recovery potential or long-term process viability.

This distinction is important not only for numerical comparison but also for interpreting treatment mechanisms and recovery pathways. Y. Ren et al. [27] noted that aqueous phosphorus may occur as orthophosphate, polyphosphate, organically bound phosphorus, and phosphite, and that

non-orthophosphate species may require conversion to reactive orthophosphate before effective removal or recovery. Accordingly, concentrations reported as TP, orthophosphate, $\text{PO}_4\text{-P}$, or PO_4^{3-} should not be treated as directly interchangeable, because they may represent different phosphorus fractions and analytical bases. Comparisons among technologies should therefore retain both the phosphorus species and the mass basis reported in the original study.

Technology-Specific Pathways and Emerging Hybrid Directions

Among the classified technologies, adsorption is widely regarded as one of the most promising approaches for phosphorus removal and recovery. It has been widely investigated as an effective technique for phosphorus removal, particularly at low concentrations. In this process, phosphate ions are captured by adsorbent materials through physicochemical interactions at the solid-liquid interface. Early studies focused on conventional materials such as activated carbon, metal oxides, and mineral-based adsorbents, which offer high surface areas and abundant active sites. More recently, research has shifted toward advanced materials, including modified biochars, LDH, metal-organic frameworks, and

nanostructured adsorbents, to enhance adsorption capacity and treatment efficiency [42, 43]. Current efforts also focus on developing materials capable of achieving high removal performance at low phosphorus concentrations while maintaining rapid kinetics and reusability [29, 30, 44].

Biological phosphorus removal, particularly EBPR, represents an established strategy for nutrient control in municipal wastewater treatment. EBPR relies primarily on polyphosphate-accumulating organisms (PAOs), which cycle phosphorus through anaerobic and aerobic metabolic stages and enable intracellular phosphorus accumulation [45–48]. As schematically illustrated in Fig. 4(b), phosphorus is ultimately concentrated within phosphorus-rich biomass, which must be separated and generally requires additional downstream processing when phosphorus recovery is intended. Microalgae-based systems have also been investigated for nutrient capture and subsequent reuse of phosphorus-enriched biomass [49]. Despite its established application, EBPR is fundamentally dependent on microbial activity and therefore remains sensitive to influent composition, carbon availability, temperature, pH, and operating conditions. Moreover, biologically captured phosphorus is primarily accumulated within biomass, meaning that effective removal from the aqueous phase does not directly yield a concentrated or readily recoverable phosphorus stream. Additional downstream processing is generally required when phosphorus recovery, rather than removal alone, is the treatment objective.

These limitations provide an important rationale for the development of electro-assisted and hybrid separation systems. Unlike EBPR, electrically driven processes can provide direct control over ion transport and separation via the applied electric potential, enabling selective phosphorus capture, controlled desorption, and enrichment of phosphorus in a smaller-volume stream. When coupled with ion-exchange membranes, selective electrodes, or complementary biological processes, electro-assisted systems can therefore separate the functions of bulk wastewater treatment, phosphorus polishing, and resource recovery. Because the present review focuses on emerging adsorption and electro-assisted hybrid technologies, biological processes are considered here primarily as a conventional benchmark and potential components of integrated treatment systems, rather than being reviewed in detail. For example, coupling biological treatment with adsorption or electro-assisted separation can distribute treatment functions across different process stages and potentially improve phosphorus polishing and subsequent recovery [50].

As an established benchmark, chemical precipitation remains widely used for phosphorus removal, particularly for treating relatively high phosphorus loads. The process involves adding aluminum-, calcium-, or iron-based salts

that react with phosphate ions to form insoluble phosphate compounds, which are subsequently separated by sedimentation or filtration. Chemical precipitation is particularly suitable for treating large wastewater volumes and can achieve high phosphorus removal efficiencies; however, it generally transfers phosphorus into a sludge or precipitated phase, which may require additional processing for subsequent recovery [51–53].

Another advanced approach, applicable either independently or in combination with other technologies, is membrane separation. Membrane technologies can remove phosphorus through size exclusion, charge interactions, or selective separation mechanisms, depending on the membrane type. Membrane processes can also be integrated with other treatment systems to enhance reliability and reduce operational energy consumption. Compared with conventional treatment technologies, membrane systems often exhibit higher selectivity and can achieve very low phosphorus concentrations in treated effluents, making them particularly suitable for advanced wastewater treatment applications [54, 55]. One promising application of membrane technology in phosphorus removal is the use of membranes as electrodes in MCDI systems. MCDI is an emerging electrochemical technology with significant potential for phosphorus removal and recovery from aqueous environments, particularly wastewater [15, 34].

The limitations of conventional biological and chemical treatment become particularly important when the treatment objective shifts from bulk phosphorus removal to selective polishing and resource recovery. EBPR can achieve effective phosphorus removal but relies on biologically mediated uptake and subsequent handling of phosphorus-rich biomass, whereas chemical precipitation often transfers phosphorus into sludge or mineral phases that may require additional processing for recovery. Electro-assisted technologies provide a fundamentally different pathway by applying an electric potential to control ion transport, adsorption, and release. In CDI and MCDI, this enables reversible phosphate electrosorption and electrically controlled regeneration, while ion-exchange membranes in MCDI can reduce co-ion effects and improve separation performance [56]. These characteristics create opportunities not only for phosphorus removal but also for its enrichment into a recoverable stream, although selectivity, competing ions, energy demand, and scale-up remain important limitations [15, 34]. Within this context, electro-assisted and hybrid systems are considered emerging pathways that complement rather than simply replace established biological and chemical treatment processes.

Despite the availability of multiple phosphorus removal technologies, increasing research attention has focused on adsorption-based systems due to their simplicity, high

efficiency at low phosphorus concentrations, and potential for resource recovery. In addition, adsorption offers significant flexibility in material design, enabling the incorporation of functional groups and metal active sites to enhance selectivity and performance. These characteristics, together with advances in composite materials, have positioned adsorption as a key focus in recent studies. Therefore, a more detailed examination of adsorption materials, mechanisms, and performance is presented in the following section.

Adsorption

Overview, Material Development, and Mechanistic Pathways

The adsorption technique has attracted significant interest in recent years due to its affordability and ease of use in phosphorus removal and recovery systems, making it a promising alternative to other techniques [57]. Despite these advantages, several inherent limitations of currently available adsorbents, such as low regeneration efficiency, high production costs, limited selectivity, and a progressive decline in adsorption capacity, continue to restrict their long-term, large-scale implementation. In addition, performance deterioration under realistic wastewater conditions, driven by competing ions and fluctuating pH, remains a critical barrier to practical deployment, requiring materials with improved selectivity and chemical stability. Taken together, these challenges emphasize the need for developing adsorbents that combine high adsorption efficiency with long-term stability and effective regeneration under realistic conditions [9, 41].

A wide range of materials with diverse origins, either organic or inorganic, have been explored as phosphorus adsorbents, including biochar [58], chitosan [59], agricultural waste-based adsorbents [60], metal oxide [61], fly ash [62], zeolite [63], and chemically modified clays [64]. In recent years, this classification has been extended to include advanced composite and hybrid adsorbents, such as metal-functionalized biochars, LDHs, MOFs, and graphene-based materials, which exhibit enhanced adsorption performance due to synergistic interactions among their structural components [65, 66]. For example, LDH-biochar composites and lanthanum-modified systems have demonstrated significantly improved adsorption capacity and stability, driven by the combined effects of interlayer anion exchange and strong metal-phosphate binding [67]. In particular, LDH-biochar systems have shown improved pH adaptability and sustained phosphorus release behavior, supporting their dual role in adsorption and resource recovery applications [67]. These developments collectively reflect a transition from single-component materials toward engineered

multifunctional systems designed to overcome the limitations of conventional adsorbents.

Based on their origin and composition, phosphorus adsorbents can be broadly classified into two main categories: organic and inorganic materials, each of which can be further subdivided into several groups, as illustrated in Fig. 5(a). Organic adsorbents are mainly derived from biomass, natural polymers, and agricultural by-products, and are valued for their low cost, renewability, and environmental compatibility [68, 69]. In contrast, inorganic adsorbents, such as metal oxides, hydroxides, and clay-based minerals, generally exhibit stronger chemical affinity toward phosphate ions and often provide higher adsorption capacities [70]. However, the distinction between these categories has become less pronounced with the emergence of composite materials that integrate organic matrices with inorganic active sites, highlighting the increasing importance of hybrid material design. This shift is not only aimed at improving adsorption capacity, but also at enhancing structural stability, selectivity, and regeneration performance under complex environmental conditions [20].

Recent studies indicate that LDHs, natural materials, and metal oxides remain among the most extensively investigated classes of phosphate adsorbents [71], while composite systems are increasingly explored due to their ability to combine multiple adsorption mechanisms and enhance performance [72]. In this context, understanding the underlying adsorption mechanisms becomes essential for interpreting the performance of these materials and guiding further material design.

From a mechanistic standpoint, phosphorus adsorption involves several pathways, including electrostatic attraction, ligand exchange, surface complexation, ion exchange, hydrogen bonding, and, in some cases, surface precipitation. In practice, these mechanisms often occur simultaneously rather than independently, and their relative contributions vary depending on material composition and environmental conditions [29, 73]. Experimental and spectroscopic analyses further demonstrate that chemisorption pathways, particularly ligand exchange and surface complexation, dominate phosphate uptake in advanced adsorbents, with contributions exceeding 70–80% in MOF-based systems [74]. The diversity of adsorption pathways in different material systems is illustrated in Fig. 5(b–e), which compares the mechanisms of phosphorus capture in biochar-based composites, MOFs, and LDHs, including electrostatic interaction, ion exchange, ligand exchange, hydrogen bonding, and coordination with metal active sites [72]. While this multi-mechanistic behavior is advantageous for enhancing adsorption efficiency, it also complicates predicting and optimizing performance under variable environmental conditions.

The relative importance of these mechanisms differs across material classes. For example, in composite biochar-based

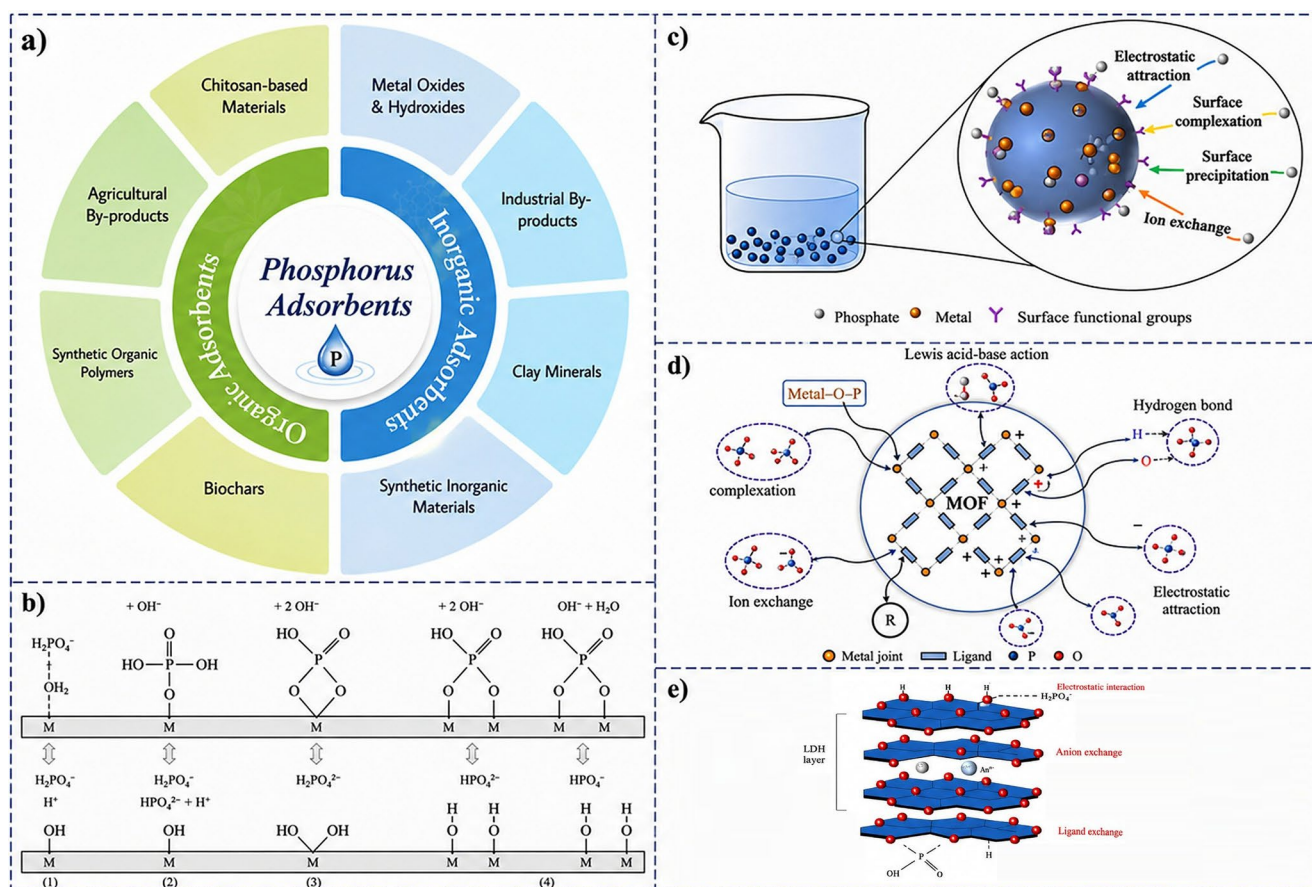


Fig. 5 (a) Phosphorus adsorbents classifications, (b) Phosphate adsorption mechanisms on metal-based adsorbents [30], (c-e) mechanisms of phosphorus capture by advanced composite adsorbents: (c) biochar-based composites, (d) MOFs, and (e) LDH [72]

systems, phosphorus capture may occur via electrostatic attraction, ion exchange, surface complexation, and surface precipitation, particularly after incorporating active metal species onto the carbon surface. In MOF-based materials, phosphate uptake is commonly associated with Lewis acid-base interactions, electrostatic attraction, hydrogen bonding, ion exchange, and coordination between phosphate species and unsaturated metal sites. Notably, ligand exchange followed by inner-sphere complexation has been identified as the dominant reaction sequence in Ce-based MOF systems, reinforcing the importance of strong chemical bonding in phosphate removal. In LDH systems, the layered structure facilitates electrostatic interaction, interlayer anion exchange, and ligand exchange at hydroxylated metal centers, which together contribute to efficient phosphate removal [72, 74–76]. This variability highlights that adsorption performance is highly material-specific and must be evaluated in relation to both structural characteristics and operating conditions rather than generalized across adsorbent categories.

The adsorption process is strongly influenced by solution chemistry, particularly pH, phosphate speciation, and

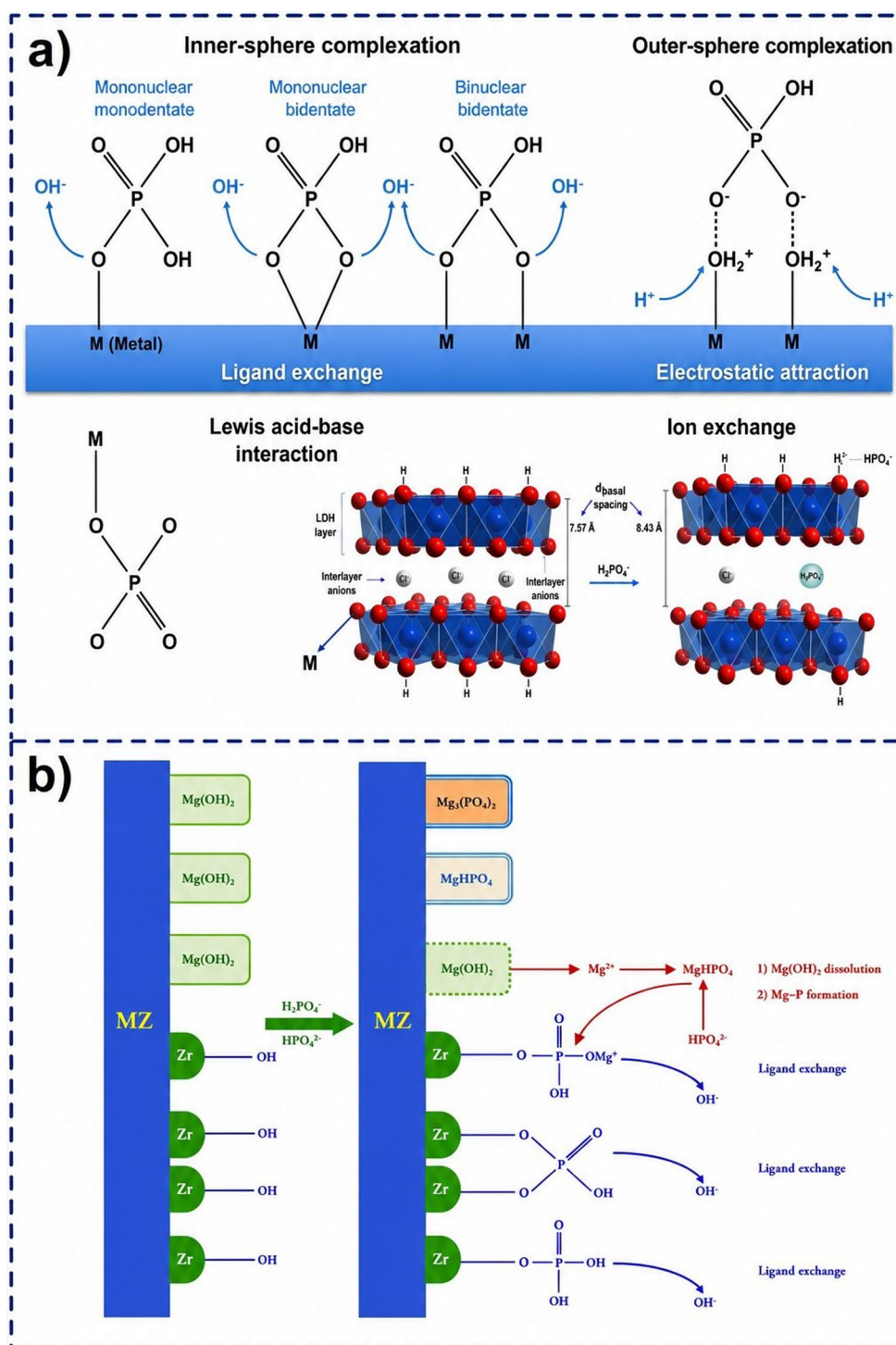
the presence of competing ions. For instance, in iron-magnesium-functionalized systems, adsorption is favored under acidic conditions due to increased surface protonation and electrostatic attraction to $H_2PO_4^-$ species, whereas adsorption efficiency decreases at higher pH due to surface deprotonation and electrostatic repulsion. Furthermore, competitive ions such as CO_3^{2-} can significantly inhibit phosphate adsorption, whereas Ca^{2+} and Mg^{2+} can enhance removal via precipitation [77]. In multi-component systems, selective adsorption is governed by differences in binding affinity, where Fe-based active sites preferentially form inner-sphere complexes with phosphate, while competing ions interact through weaker mechanisms such as electrostatic attraction or hydrogen bonding [78]. Recent studies further confirm that adsorbents with strong inner-sphere complexation (e.g., La- or Fe-based materials) exhibit lower sensitivity to ionic strength but may suffer from reduced regeneration efficiency [79]. Therefore, the strong dependence of adsorption performance on environmental conditions represents a key limitation for real wastewater applications.

Mechanistic Insights: Surface Complexation and Precipitation

In metal-based adsorbents, phosphate uptake is generally governed by a combination of electrostatic attraction and ligand exchange processes. While electrostatic interactions assist in bringing negatively charged phosphate species toward positively charged surfaces, ligand exchange is

typically the dominant mechanism, leading to the formation of inner-sphere complexes [30, 80]. As shown in Fig. 6(a), these inner-sphere complexes involve direct coordination between phosphate oxygen atoms and surface metal centers, forming monodentate or bidentate structures depending on the binding configuration [80, 81]. This strong inner-sphere binding is widely reported in lanthanum- and iron-based systems and is considered a key factor responsible for their

Fig. 6 (a) Schematic of phosphate adsorption mechanisms on metal-based adsorbents, highlighting inner-sphere (monodentate/bidentate) and outer-sphere complexation [80], (b) Phosphate removal mechanisms in the MZ composite: ligand exchange and Mg-P precipitation [80]



high adsorption capacity and selectivity [79]. In contrast, outer-sphere complexation relies on weaker electrostatic interactions without direct bonding. The dominance of inner-sphere binding is further supported by its relatively low sensitivity to ionic strength and competing anions. However, although such strong binding enhances adsorption stability and selectivity, it can hinder phosphate release during regeneration. Consequently, this introduces a fundamental trade-off between adsorption capacity and regenerability, which remains a key challenge in the design of reusable phosphate adsorbents [82].

In addition to surface complexation, phosphate removal in metal-rich systems can also occur via precipitation. As shown in Fig. 6(b) for the MZ ($\text{Mg}(\text{OH})_2/\text{ZrO}_2$) composite, Mg-containing phases can dissolve and subsequently react with phosphate to form stable Mg-P compounds, while Zr-OH sites contribute through ligand exchange. Similar precipitation-assisted mechanisms have also been observed in calcium- and magnesium-modified biochars, where the formation of stable phosphate minerals contributes to high removal efficiency but may limit phosphorus recovery. This combined mechanism, involving both adsorption and precipitation, enhances phosphorus removal and promotes long-term immobilization [80, 83, 84]. Although precipitation is beneficial for permanent phosphorus removal, it may limit phosphorus recovery, an important consideration in circular-economy-based wastewater treatment. At the nanoscale, adsorption behavior is also influenced by structural properties such as pore size and degree of oxidation. For example, in three-dimensional reduced graphene oxide materials, less compact structures with higher oxidation levels provide more accessible functional groups and suitable pore environments, facilitating ion clustering and stronger adsorption interactions [85]. These observations further highlight that nanoscale structural engineering plays a critical role in improving adsorption performance.

Emerging Adsorbent Development: Material Advancements and Practical Challenges

The development of adsorbents for phosphorus removal has progressively shifted from conventional, naturally available materials to engineered, multifunctional adsorbents designed to improve phosphate affinity, selectivity, regeneration, and stability. Early adsorbents, including activated carbon, zeolites, clays, and unmodified biochars, offered relatively simple, low-cost platforms but were often limited by moderate adsorption capacity and poor phosphate selectivity. Subsequent material development therefore focused on surface modification and incorporation of phosphate-affinitive metal species, particularly Fe, Al, Mg, Ca, La, and Zr, leading to the development of modified biochars,

metal oxides/hydroxides, LDHs, and composite adsorbents. More recently, highly tunable porous materials such as MOFs and other hybrid or nanostructured adsorbents have enabled greater control over surface chemistry, pore structure, active-site distribution, and phosphate-binding mechanisms. This evolution reflects a broader transition from maximizing adsorption capacity alone toward designing materials that combine high affinity and selectivity with regenerability, structural stability, and applicability under realistic wastewater conditions [86, 87].

Direct ranking of the adsorbents based solely on removal percentage or maximum adsorption capacity should therefore be avoided, particularly when initial phosphorus concentration, adsorbent dosage, pH, contact time, and water matrix differ substantially among studies. A critical evaluation of the data presented in Table 4 reveals several important trends in phosphorus adsorption performance. First, metal-modified and composite adsorbents, particularly those incorporating lanthanum, calcium, or iron, generally exhibit higher adsorption capacities compared to unmodified materials, indicating the significant role of metal active sites in enhancing phosphate binding. Second, adsorption capacity is strongly influenced by the initial phosphorus concentration and experimental conditions, which limit direct comparability across studies. Lastly, materials with high removal efficiency at low phosphorus concentrations are particularly relevant for tertiary wastewater treatment applications, where stringent discharge limits must be met. Furthermore, some materials exhibit exceptionally high adsorption capacities under controlled laboratory conditions, but their practical applicability may be limited by factors such as regeneration efficiency, stability, and sensitivity to pH and competing ions. Overall, these findings highlight a clear gap between laboratory-scale performance and real-world applicability [87, 117].

Biochar-based adsorbents derived from agricultural or organic wastes have attracted increasing attention for their cost-effectiveness and environmental sustainability. Their properties can be tailored through controlled pyrolysis conditions, activation processes, and chemical modification, which significantly influence surface area, pore structure, and functional group distribution. Fundamentally, high P recovery efficiency using biochar can be achieved through effective waste management and recycling processes. Nevertheless, pristine biochar often exhibits limited adsorption efficiency due to its negatively charged surface and inherent phosphorus content, which may even lead to phosphorus release under certain conditions [118–120]. To address these limitations, biochar is frequently modified with metals such as Ca, Mg, Fe, La, and Al. These modifications enhance adsorption performance by introducing positively charged active sites, facilitating ligand exchange reactions, and

Table 4 Adsorption characteristics data on phosphorus or phosphate adsorption from aqueous solutions according to recent publications

Material Class	Adsorbent	Water Matrix	Initial P/PO_4^{3-} Concentration (mg/L)	pH	Adsorbent Dose (mg/L)	Adsorption Time (h)	Removal Efficiency (%)	Adsorption Capacity (mg/g)	Regeneration/Reuse No. of Cycles	Key Insight	Ref.
Bio-based/ Waste-derived Adsorbents	Eggshell-Modified Biochar (EGBC)	Synthetic KH_2PO_4 solution; simulated wastewater in column test	200 (P)	7 (baseline); effective over pH 5–12	700	37.5	~99	288.83	4 cycles; 65% recovery and 121 mg/g after cycle 4	Ca-mediated precipitation dominates adsorption	[88]
	Banana peel and eggshell biocomposite (BPES 1:2)	Synthetic P solution; tertiary-treated domestic wastewater also tested	200 (P)	NR	1000 (removal test); 500 (isotherm)	2	88	214.1	NR; spent P-loaded material evaluated for fertilizer use rather than regeneration	Ca-rich phases enable precipitation pathways	[9]
	Chlorella vulgaris	Synthetic phosphate solution; agricultural wastewater also tested	20 (PO_4)	7	80% algae mass ratio (reported basis)	24	~80	250	NR	Biosorption coupled with surface binding	[89]
	Calcium-rich biochar	Synthetic P solution; treated urban wastewater also tested	200 (P)	2–10 tested; highest uptake at pH 10	1000	20	-	158.8	NR; P-loaded biochar proposed for direct agricultural reuse	Ca-driven precipitation and complexation	[90]
	ZnAl-LDH modified plane trees' bark biochar	Synthetic phosphate solution	50 (PO_4)	NR	400	2	93	130.1	2 adsorption-desorption rounds; strong decline in round 2; ~57–61% first desorption with Na_2CO_3	Anion exchange and surface complexation	[91]
	La-C-550 (lanthanum-coated sludge biochar)	Synthetic KH_2PO_4 solution; actual secondary effluent also tested	10 (P)	6.6 ± 0.2 (baseline); stable over pH 3–7	100	6	-	76.4	4 cycles; 45.2 mg-P/g in cycle 4 (>60% of initial)	La active sites enhance ligand exchange	[92]
	La-BRWs (lanthanum-modified biorefinery residue of wheat straw)	Synthetic phosphate solution	162.29 (P)	1.01–10.83 tested	10,000	0.33	>99.4	55.74	5 cycles; >99% regeneration efficiency with 5 M NaOH	La incorporation improves binding affinity	[93]
	Boehmite-modified carbon composite (moldy corn)	Livestock wastewater	32 (P)	6.47 native; pH 2–10 examined	2000	12	21.75	40.98	6 cycles	Weak surface complexation limits capacity	[94]

Table 4 (continued)

Material Class	Adsorbent	Water Matrix	Initial P/PO_4^{3-} Concentration (mg/L)	pH	Adsorbent Dose (mg/L)	Adsorption Time (h)	Removal Efficiency (%)	Adsorption Capacity (mg/g)	Regeneration/Reuse No. of Cycles	Key Insight	Ref.
Metal-based Adsorbents	Lanthanum-modified reinforced composite	Synthetic phosphate solution	20 (P)	6 optimum; >90% removal over pH 4–10	12,000	48	90	20.395	3 cycles tested; ~2 practical reuses; cycle 3 removal 62.37% (2.5 M NaOH, 24 h desorption)	La-induced complexation enhances uptake	[95]
	Sponge iron mixed with biochar	Artificial low-C/N wastewater in bench-scale constructed wetland; domestic sewage used for start-up	0.06 (P)	NR (monitored during operation)	1:9 sponge iron: biochar filler ratio (reported basis)	-	92.1	-	NR (fixed constructed-wetland substrate)	Fe active sites enable low-level adsorption	[96]
	Thermally activated mandarin peel waste	Synthetic phosphate solution (DI water)	10 (PO_4)	5.9	750	1	99	9.0	NR	Surface adsorption with limited active sites	[97]
	Calcium silicate hydrate	Synthetic water; household laundry wastewater; RO concentrate	100 (PO_4)	NR	NR (5-cm CSH column)	20	99.94	308.5	NR	Ca-induced precipitation governs uptake	[98]
	Lanthanum hydroxide	Synthetic phosphate solution; natural lake water	30 (PO_4)	7 ± 0.2	100	2	~100	308	5	Inner-sphere La-phosphate complexation	[99]
	Sepiolite-supported MgO	Synthetic nutrient solution (distilled water)	20 (mmol/L of P)	7.0	600	2	86	188.49	NR	Mg dissolution enables precipitation	[100]
	Lanthanum-modified sludge-based biochar	Synthetic P solution; fixed-bed applicability test	15 (PO_4)	3.0	1300	24	99.06	152.77	6 (76.6% retained)	La-driven ligand exchange dominates	[101]
	LZ (lanthanum chloride-modified zeolite)	Synthetic phosphate solution; treated wastewater effluent	10 (P)	6	300	24	-	124.38	5 (>70% retained)	Ion exchange and La complexation	[79]
	Al-modified hydrochar	Synthetic KH_2PO_4 solution; simulated wastewater	500 (PO_4)	7 (optimum)	3333	6	76.27	114.40	NR	Al-phosphate surface complexation	[102]
	ZnAlLa-LDH	Synthetic phosphate solution; real wastewater	10 (P)	7 ± 0.1	100	12	~100	105.42	5 (>73% retained)	Combined ion exchange and ligand exchange	[103]
	Fe-CLCAB	Aqueous phosphate solution; pond, municipal, and groundwater	10 to 200 (PO_4)	5	NR (500 beads/25 mL)	5	94	73.13	6 (42% of original efficiency)	Fe-based surface complexation and redox activity	[104]

Table 4 (continued)

Material Class	Adsorbent	Water Matrix	Initial P/PO_4^{3-} Concentration (mg/L)	pH	Adsorbent Dose (mg/L)	Adsorption Time (h)	Removal Efficiency (%)	Adsorption Capacity (mg/g)	Regeneration/Reuse No. of Cycles	Key Insight	Ref.
MOFs and Advanced Porous Materials	Fe-Mn oxide/mulberry branch biochar composite	Synthetic phosphate solution; river, agricultural, municipal, and pharmaceutical wastewater	30 (P)	3.0	NR (2-cm fixed bed)	2.1	98.1	22.11	4 (> 72% removal retained)	Metal oxide surface complexation dominates	[105]
	La-MgAl-LDH/BC	Synthetic phosphate solution; river water	50 (PO_4)	6	400	1	-	249.3	4 (63.5% uptake retained)	Synergistic metal-site complexation	[106]
	ZIF-8 modified with 2-phenyl imidazole	Synthetic phosphate solution	150 (PO_4)	Neutral (~7)	200	4	-	162.93	5 (101.81 mg/g after cycle 5)	Functional groups enhance coordination sites	[107]
	MOF-76 (Ce)	Synthetic KH_2PO_4 solution	20 (PO_4)	5.5	NR (1 mg per test)	3	-	72.97	NR	Ce-phosphate coordination dominates	[108]
	Ce-Fum700-BC-CMC100 (immobilized Ce-MOF composite)	Synthetic phosphate solution; five real waters (lake, paddy field, pond, tap water, groundwater)	5 (P)	6.8 (validation test)	1.5 (equilibrium)	1.5	99.93	67.35	7 (>90% removal after 5 cycles; ~80% after 7)	Ligand exchange with improved stability	[74]
Minerals/Clays	CeO_2	Synthetic KH_2PO_4 solution	20 (PO_4)	5.5	NR (1 mg per test)	3	-	55.71	NR	Surface hydroxyl complexation mechanism	[108]
	Fe/AlPMt (iron-modified montmorillonite/ Al_2O_3 composite)	Synthetic phosphate solution; lake water	20 (P)	3.0 (kinetic tests; effective pH 3–6)	1000 (batch); 300 in lake-water test	4	-	51.41	5 (95.8% adsorption retained)	Fe/Al sites enable ligand exchange	[109]
	Kaolin	Synthetic storm-water (DI-water phosphate solution)	0.5 to 12 (PO_4)	NR (not adjusted)	50,000	24	~100 to 56	0.141	NR	Weak physisorption dominates	[110]
	Bentonite	Synthetic storm-water (DI-water phosphate solution)	0.5 to 12 (PO_4)	NR (not adjusted)	50,000	24	40 to 12	0.033	NR	Limited ion exchange capacity	[110]
Polymer/Hydrogel Systems	Chitosan-based yttrium hydrogel	Synthetic phosphate solution	50 (P)	6	1000 (0.1% solid-liquid ratio)	2	80	62.36	7 (>80% capacity retained; 10 cycles tested)	Metal-assisted coordination binding	[111]
	MB: PVDF membrane	Synthetic phosphorus feed; humic-acid interference also tested	5.8 (P)	6–7 (reported basis)	Membrane basis; NR	24	48.9	53.5 (mg/m ²)	NR	Surface adsorption on immobilized phase	[112]

Table 4 (continued)

Material Class	Adsorbent	Water Matrix	Initial P/PO_4^{3-} Concentration (mg/L)	pH	Adsorbent Dose (mg/L)	Adsorption Time (h)	Removal Efficiency (%)	Adsorption Capacity (mg/g)	Regeneration/Reuse No. of Cycles	Key Insight	Ref.
Industrial/Waste-derived Mineral Adsorbents	PVA/SA matrix double network hydrogels	DI, tap, pond, lake, and river water	200 (P)	6	1000	2	12.7	38.75	8 (~9.67 mg/g after cycle 8)	Limited functional groups for binding	[113]
	ZnO/ZnFe-LDHs/alginate hydrogel	Synthetic phosphate solution	48 (PO_4)	7.0	600	10	90	17.06	5 (> 90% removal retained; 96.81% P recovery)	LDH-driven ion exchange and precipitation	[114]
	Medium-MW Chitosan	Synthetic storm-water (DI-water phosphate solution)	0.5 to 12 (PO_4)	NR (not adjusted)	50,000	24	84% to 31	0.083	NR	Adsorption influenced by polymer structure	[110]
	Low-MW Chitosan	Synthetic storm-water (DI-water phosphate solution)	0.5 to 12 (PO_4)	NR (not adjusted)	50,000	24	77% to 19	0.049	NR	Low density of active binding sites	[110]
	High-MW Chitosan	Synthetic storm-water (DI-water phosphate solution)	0.5 to 12 (PO_4)	NR (not adjusted)	50,000	24	55% to 18	0.051	NR	Diffusion-limited adsorption behavior	[110]
Industrial/Waste-derived Mineral Adsorbents	Foundry dust-based geopolymer	Synthetic N/P solution; real swine wastewater in column test	10 (P)	9	200	12	90.92	175.09	NR	Surface precipitation and ion exchange	[115]
	High-silica Iron tailing adsorbent	Synthetic KH_2PO_4 solution	1.1 (P)	7.5 ± 0.1	NR ($\approx$ 1-cm particles; mass not reported in accessible text)	2	68.13	0.51	NR	Limited active sites for binding	[116]

NR means not reported in the original study. Phosphorus concentrations and adsorption capacities retain the basis used in the source study unless otherwise stated. Values originally reported in mg/kg (Ref [110]), were converted to mg/g to match the table heading. For membrane/column studies, technology-specific operating information is reported where a conventional batch adsorbent dose or contact time is not applicable

promoting precipitation mechanisms. However, the reliance on externally sourced metal precursors may increase production costs and partially offset the sustainability benefits of biochar-based systems [9, 76, 121].

The incorporation of calcium-rich materials, such as eggshells, into biochar has been shown to significantly improve phosphorus removal efficiency. For instance, a banana peel and eggshell biocomposite (1:2 ratio) demonstrated high adsorption efficiency in tertiary-treated wastewater [9]. This improvement is attributed to the transformation of CaCO_3 into CaO and Ca(OH)_2 , which enhances phosphate removal through surface complexation and precipitation mechanisms [9, 122]. While effective, this mechanism may form stable calcium-phosphate compounds, potentially limiting phosphorus recovery and adsorbent reusability.

Composite systems such as sponge iron-biochar [96], calcium-rich biochar [90], La-doped LDH/biochar composite [106], and Fe-Mn oxide/mulberry branch biochar composites [105] demonstrate synergistic effects that enhance adsorption performance. For example, sponge iron-biochar systems exhibit high removal efficiency due to enhanced micro-electrolysis and increased availability of active binding sites [96]. Despite these advantages, such composites often exhibit a gradual decline in performance due to active-site saturation, structural degradation, and interference from competing ions, thereby limiting their long-term applicability in real wastewater systems. Furthermore, studies on composite materials such as La-MgAl-LDH/BC indicate that adsorption efficiency decreases over repeated regeneration cycles due to competitive anions and saturation of active sites [106]. Similarly, Fe-Mn biochar composites show reduced performance at elevated pH and after multiple reuse cycles [105].

Beyond biochar-based systems, several synthesized materials, including hydrogels, mineral-based adsorbents, and nanocomposites, have been developed for phosphorus removal. For example, Boehmite-modified carbon composites and PVA/SA hydrogels demonstrate promising adsorption performance but face challenges related to by-product generation, complex synthesis, and potential leaching risks [94, 113]. Similarly, mineral-based adsorbents such as MgO-modified geopolymers and mixed matrix membranes show high efficiency but are strongly influenced by pH and the presence of organic matter [115]. These limitations suggest that high adsorption capacity alone is insufficient; material robustness and environmental safety must also be considered in the design of adsorbents.

Advanced materials such as MOFs and nanocomposites represent a new generation of phosphorus adsorbents with tunable structures and high surface areas. For instance, sepiolite-supported MgO and modified ZIF-8 materials

demonstrate enhanced adsorption capacity [100, 107]. However, these materials face challenges related to scalability, regeneration, and potential secondary contamination. Thus, while advanced materials offer superior performance, their practical implementation is often constrained by economic and environmental considerations.

To enable a more meaningful comparison among adsorbents, Table 4 summarizes not only removal efficiency and adsorption capacity but also the experimental conditions under which these values were obtained, including the water matrix, initial phosphorus concentration, pH, adsorbent dosage, contact time, and, where reported, regeneration performance. These parameters are important because apparently similar removal efficiencies or adsorption capacities may be obtained under substantially different experimental conditions.

A further limitation when comparing adsorption studies is that high phosphate uptake does not necessarily indicate effective phosphorus recovery. In strongly binding metal-based adsorbents, for example, inner-sphere complexation or phosphate precipitation can enhance removal capacity and resistance to competing ions, but the same interactions may hinder desorption and regeneration of the adsorbent. Conversely, materials that permit easier phosphate release may offer better recovery potential despite exhibiting lower maximum adsorption capacities. The practical value of an adsorbent should therefore be assessed by balancing adsorption capacity, selectivity, regeneration efficiency, stability over repeated cycles, and the concentration and usability of the phosphorus-containing regenerant.

The integration of artificial intelligence (AI) in adsorbent development has recently emerged as a promising approach for optimizing synthesis conditions and improving adsorption performance [102]. However, this approach remains in its early stages, and its practical impact on large-scale applications has yet to be fully established. Biological approaches, such as the use of microalgae (e.g., *Chlorella vulgaris*), provide an alternative pathway for phosphorus removal and recovery [89]. These systems offer advantages in terms of nutrient recycling and sustainability. Nevertheless, their slower kinetics, operational complexity, and sensitivity to environmental conditions limit their direct comparison with adsorption-based systems in high-throughput wastewater treatment applications.

Despite the promising performance of various adsorbents, several challenges remain, including potential metal leaching, high energy consumption during activation processes, and limited mechanical and chemical stability under industrial conditions [97, 102]. Addressing these challenges is essential to ensure that adsorption technologies do not introduce secondary environmental risks while maintaining high efficiency.

While adsorption technologies have demonstrated considerable promise for phosphorus removal, their practical deployment remains constrained by scalability, cost, and long-term operational performance. Addressing these limitations requires a shift toward application-oriented research that bridges the gap between laboratory-scale innovation and real-world implementation. In particular, the development of multifunctional adsorbents that selectively capture phosphorus in the presence of competing ions is essential. Greater emphasis should also be placed on low-cost, waste-derived materials to enhance economic feasibility and sustainability.

Equally important is aligning material design with process-level requirements. Advanced surface modification strategies should focus on maintaining performance under complex wastewater conditions while minimizing fouling and improving regeneration efficiency. Furthermore, pilot-scale validation under realistic and dynamic conditions is necessary to assess long-term stability. Finally, integrating adsorption with complementary biological or membrane-based processes represents a promising pathway toward more robust and adaptable phosphorus removal systems.

Overall, the progression of phosphorus adsorbents demonstrates that increasing adsorption capacity alone is no longer sufficient to define material advancement. Although engineered biochars, metal-based adsorbents, LDHs, MOFs, and hybrid materials have substantially improved phosphate affinity and adsorption performance, their practical value increasingly depends on selectivity in multi-ion systems, structural stability, regeneration over repeated cycles, and performance in real wastewater. Future material development should therefore move beyond capacity-driven optimization toward adsorbents designed for reversible phosphorus capture and recovery under continuous and realistic operating conditions. Such a shift is essential for translating high laboratory-scale adsorption performance into practical phosphorus-management systems.

Electro-Assisted Membrane Hybrid Technologies

Membrane-based processes provide an important transition from conventional phosphorus treatment to more selective, recovery-oriented systems. Unlike EBPR, where phosphorus removal depends on microbial uptake and subsequent biomass handling, membrane and electro-assisted processes can directly manipulate phosphorus transport and concentration through size exclusion, charge interactions, selective ion transport, and electrically controlled adsorption/desorption. This distinction becomes particularly important when the objective extends beyond meeting discharge limits to

producing a concentrated phosphorus stream suitable for downstream recovery. Conventional membrane processes nevertheless remain limited by fouling, energy requirements, and insufficient phosphate selectivity in complex wastewater matrices [123–129]. These limitations have driven increasing interest in electro-assisted and hybrid membrane configurations that combine physical separation with electrochemical control, adsorption, or selective ion transport.

In wastewater treatment and water purification applications, membranes play a necessary role in microfiltration (MF), ultrafiltration (UF), nanofiltration (NF), and reverse osmosis (RO) processes [130]. One practical example is seawater desalination, which removes salts and impurities to produce potable water. In this application, UF membranes are employed to strongly reject bacteria, viruses, and suspended solids present in seawater [123]. Beyond water purification, membranes are increasingly used in wastewater treatment, particularly for the removal and recovery of P. Their primary advantage lies in their ability to separate P with exceptionally high purity and efficiency compared to alternative technologies. Membrane-based systems are emerging as advanced techniques for P recovery due to their demonstrated high nutrient rejection rates, especially for PO_4^{3-} . At the same time, the separation achieved in these systems is largely non-specific, meaning that phosphate is removed together with other anions, which introduces challenges in downstream recovery and selectivity under realistic wastewater conditions [131].

In P removal and recovery applications, membranes serve as thin, selective barriers that separate P from wastewater. However, hybrid systems such as osmotic membrane bioreactors (OMBR) and microfiltration-forward osmosis membrane bioreactors (MF-FO-MBR) have been developed beyond standalone membrane applications. These systems are designed to achieve high PO_4^{3-} and nutrient rejection while creating conditions favorable for chemical precipitation in PO_4^{3-} recovery processes. Nevertheless, even in these advanced configurations, the lack of intrinsic selectivity toward phosphate over competing anions (e.g., sulfate, nitrate, and bicarbonate) remains a key limitation, particularly under realistic wastewater conditions with high ionic strength and complex composition [131].

Furthermore, P recovered through membrane applications typically exhibits the purity required for reintegration into the agriculture industry, making it suitable for recycling and reuse. This highlights the potential of membrane technologies to contribute not only to efficient wastewater treatment but also to sustainable nutrient management [132, 133]. Membranes can be categorized by their components: polymeric membranes [134], inorganic membranes [135], composite membranes [136], biological membranes [137],

mixed matrix membranes [138], and electrospun membranes [139]. Building on these developments, recent research has increasingly shifted toward hybrid and electro-assisted configurations, where separation is coupled with adsorption, electrochemical reactions, or biological processes, enabling a more controlled and multifunctional approach to phosphorus removal and recovery [15]. In addition, recent advances in membrane material engineering have demonstrated that interfacial modification can significantly enhance phosphorus selectivity and transport behavior. For instance, MOF-interlayered forward osmosis membranes have achieved phosphorus rejection efficiencies of up to 99% while increasing water flux by approximately 57%, highlighting the potential of nanostructured interfaces to overcome traditional permeability-selectivity trade-offs [140]. This shift reflects a broader transition from purely separation-based systems toward integrated platforms capable of simultaneously removing, concentrating, and recovering phosphorus in a controllable manner [15]. These integrated systems aim to overcome key limitations of conventional membrane processes, particularly in terms of selectivity, fouling control, regeneration, and phosphorus recovery efficiency [141]. This section covers recent developments in the use of membranes for P recovery, followed by a review of process enhancements using MCDI. Different membrane methods in the recovery of P are illustrated in Fig. 7(a).

Within the broader range of membrane-based and hybrid technologies discussed in this section, electrochemical approaches to phosphorus management encompass several mechanistically distinct pathways. These can broadly be grouped into (i) electrocoagulation and electrochemically induced precipitation, where electrical input generates reactive species or local conditions that promote phosphate capture in a solid phase; (ii) electrically driven membrane separation, including electrodialysis and related ion-selective membrane processes, where phosphorus-containing ions are transported and concentrated across membranes; and (iii) electrosorption-based technologies, including CDI, MCDI, and FCDI, where ions are reversibly captured at charged electrode interfaces. These pathways differ substantially in their dominant phosphorus-control mechanism, energy requirements, regeneration behavior, concentration capability, and potential for producing a recoverable phosphorus stream.

Membrane-based Hybrid Biological Systems

Membranes have also been integrated into hybrid systems alongside other technologies to enhance the efficiency and selectivity of P removal from aqueous solutions. Examples of these complementary technologies include electrochemical methods, adsorption processes, and bioreactors. Among

these, biologically integrated membrane systems have attracted particular attention, as they combine physical separation with biologically driven nutrient transformations. The development of advanced wastewater treatment technologies focused on nutrient removal, particularly N and P, is essential to mitigate their adverse environmental impacts, such as eutrophication in aquatic ecosystems.

Advancing innovative nutrient management methods, such as the anaerobic-anoxic-oxic-anoxic membrane bioreactor ($A^2O/A+MBR$) system, CDI electrodes, and cathodic membrane filtration (CMF) reactors, is a crucial step toward more sustainable, environmentally friendly wastewater treatment processes. These cutting-edge approaches offer improved nutrient recovery and removal, paving the way for more efficient and eco-conscious water management solutions [56, 143, 144]. The $A^2O/A+MBR$ system, developed through an integrated approach, has enhanced nutrient removal capabilities by optimizing internal recirculation ratios and incorporating a post-anoxic unit. This configuration leverages the coordinated actions of microbial communities, including phosphorus-accumulating organisms (PAOs) and denitrifying glycogen-accumulating organisms (DGAOs), to achieve substantial improvements in nutrient removal efficiencies. Specifically, the system has demonstrated increases in N and P removal efficiencies of 9.9% and 16.3%, respectively. By capitalizing on microbial synergies, the $A^2O/A+MBR$ system represents a promising advance in wastewater treatment technology. More recently, biomineralization-driven anaerobic membrane systems have demonstrated phosphorus removal efficiencies as high as 96.8%, while simultaneously increasing methane production from 0.20 to 0.29 L CH_4 g^{-1} COD, indicating strong potential for integrating nutrient recovery with energy generation [145]. However, its complexity and the challenges associated with maintaining stable microbial interactions and operational conditions remain significant hurdles to its broader adoption [143]. In practice, this means that system performance is highly sensitive to operational fluctuations, particularly variations in influent composition, which can destabilize microbial communities and reduce overall efficiency [34].

Furthermore, composite functional particles were used to enhance the adsorption properties and biological activities of a bioreactor, known as a gravity-driven ceramic membrane bioreactor (GDCMBR). This application not only increases nutrient removal but also enhances the ability to ensure the safety of potable water. The main advantages of such a removal and recovery system are its cost-effectiveness and higher biological activity. However, it faces challenges, including initial flux reduction and complexity in preparing the composite particles [146]. Moreover, these systems are typically designed with a primary focus on

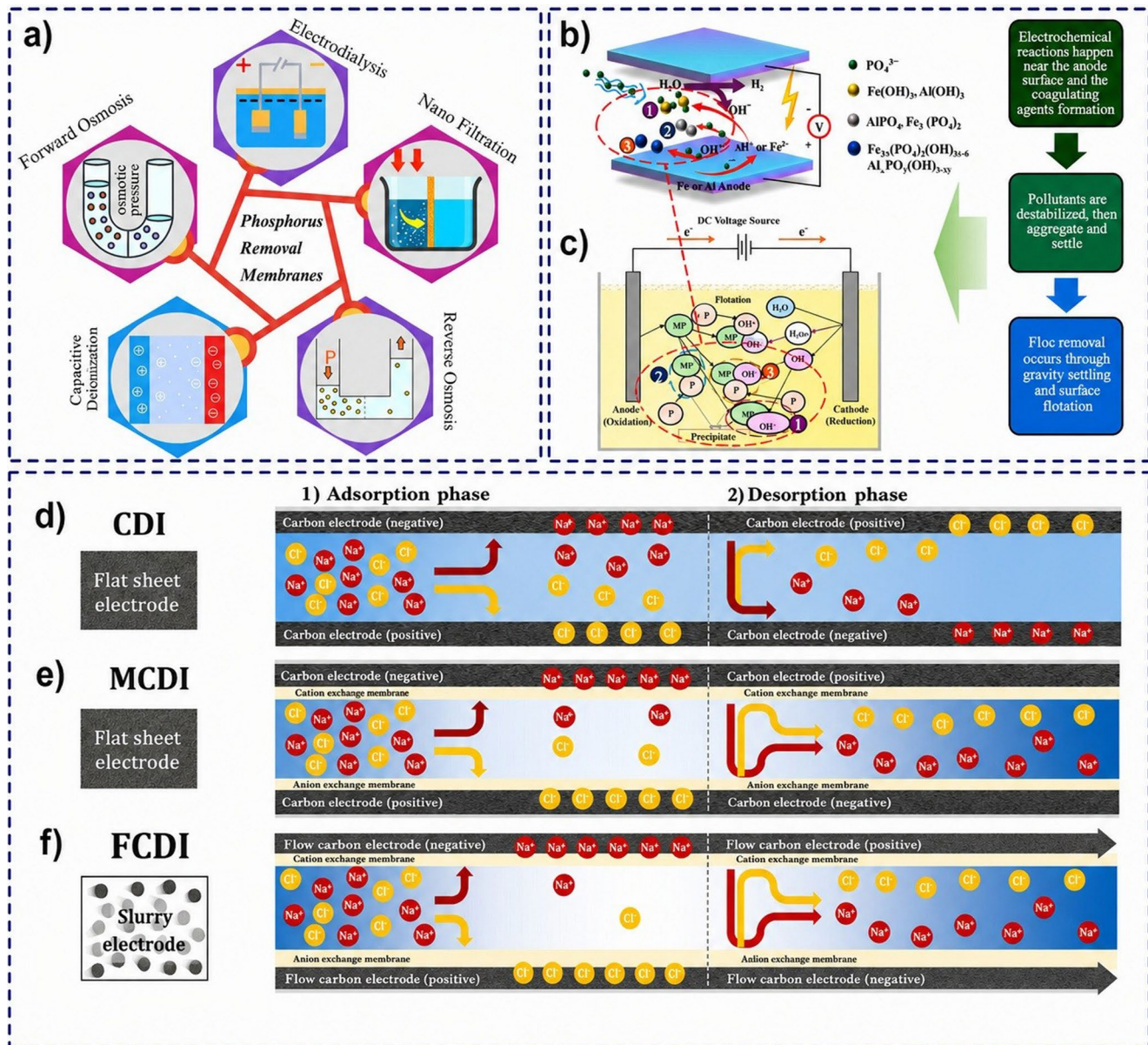


Fig. 7 (a) Major membrane techniques that have been employed in P removal, (b) Electrochemical reactions at the electrode surface leading to metal ion release and phosphate precipitation [142], (c) floc formation, aggregation, and separation via settling and flotation [142],

(d) CDI based on electro-sorption onto porous electrodes [141], (e) MCDI with improved selectivity via ion-exchange membranes [141], and (f) FCDI enabling continuous operation with slurry electrodes [141]

removal efficiency rather than selective phosphorus recovery, which limits their role in circular nutrient management strategies [15]. In parallel, the incorporation of low-cost mineral-based adsorbents into membrane bioreactors has been shown to improve phosphorus removal efficiency from approximately 35% to 74%, emphasizing the importance of cost-effective material selection for large-scale applications [147].

To improve the treatment of an aqueous solution with high concentrations of N and P, an innovative approach was

developed: an integrated reactor system that combines an anaerobic ceramic membrane bioreactor (AnCMBR) with an anaerobic fluidized bed (AFB). The system exhibited reasonable performance in pollutant removal and control of membrane fouling. Despite the fact that the integrated reactor system is complicated, it presents great nutrient recovery capabilities and efficient fouling control, which are key parameters that should convince researchers to undertake more detailed investigations into development opportunities [148].

Electrochemical-Membrane Systems for Nutrient Recovery

A different approach to removing P involves a cathodic membrane filtration (CMF) reactor system that combines electrochemical processes with membrane filtration to achieve efficient P removal. Under ideal experimental conditions, the system has demonstrated a removal efficiency of 96.2%. In the CMF reactor, the generation of a locally high pH at the cathode surface promotes the nucleation of calcium phosphate, primarily as hydroxyapatite (HAP), a stable and valuable byproduct. While CMF technology boasts low energy consumption and high removal efficiency, it faces challenges such as potential membrane fouling and operational complexity, which necessitate further research to optimize the system for real-world applications [144]. In this context, the formation of hydroxyapatite at the cathode surface can be understood as an electrochemically induced precipitation process, consistent with mechanisms reported in other electro-assisted recovery systems [149]. Building on these concepts, electroactive membrane reactors (EMRs) have recently been developed to enable simultaneous phosphonate degradation (~85%) and phosphorus recovery (~80%) within a single unit, with reported energy consumption as low as 0.019 kWh/g P, demonstrating a significant step toward process intensification [150].

Another innovative technology for the removal of N and P is bipolar membrane electrodialysis (ED-MB). This method achieves remarkable removal efficiencies of up to 96.5% for ammonium (NH_4^+) and 75.4% for PO_4^{3-} , making it highly effective. Additionally, the nutrients in aqueous solutions can be directly converted into valuable resources, such as fertilizers, aligning with sustainable resource recovery practices. Despite these advantages, ED-MB technology is burdened by high operational and maintenance costs and significant energy consumption. This limitation is not unique to ED-MB and is commonly observed across electro-driven membrane processes, particularly when moving from laboratory to continuous operation [15, 151].

Recently, polymers have become ideal candidates for various applications due to their excellent properties, including high thermal stability, mechanical strength, and outstanding transport properties. The doping method significantly enhances critical features, such as proton conductivity, water retention, and chemical resistance, but it also increases the cost of polymer production. These improvements make polymers highly suitable for use in separation processes. Membranes based on these materials can facilitate controlled permeation and retention of P species, thereby enhancing the efficiency and performance of P recovery processes [152]. A novel P removal and recovery method involves preparing membranes incorporating clay

adsorbents into a polyvinylidene fluoride (PVDF) matrix. Chemical modification of the clay adsorbents within the membrane structure significantly improved the adsorption capacity. This modification enhanced the material's physical and adsorption properties, resulting in a reduced adsorption equilibrium time of approximately 360 min and increased porosity, both of which are critical for adsorption efficiency. However, the increased porosity compromises the material's mechanical strength and raises concerns about potential fouling issues, making it less favorable for use in filtration processes [153]. This highlights a common trade-off in hybrid membrane design, where improvements in adsorption capacity may come at the expense of structural stability and long-term operability [154].

LDH membranes have emerged as a promising technology for PO_4^{3-} removal and recovery, particularly in large-scale wastewater treatment. The mechanisms underlying this technology include ion exchange, ligand exchange, and electrostatic attraction. LDH membranes have demonstrated excellent permeability, high removal efficiency, and the capacity to handle significant wastewater volumes, making them suitable for real-world applications. However, challenges such as competitive adsorption by coexisting ions and metal-ion loss during regeneration remain critical issues requiring further attention [155]. Notably, the strong affinity of LDH materials toward phosphate is attributed to their layered structure and hydroxyl-rich surfaces, which promote ion exchange and inner-sphere complexation mechanisms [154].

Taken together, these electrochemical-membrane systems demonstrate that integrating membranes with electrochemical processes can improve nutrient recovery and, to some extent, enhance selectivity. However, their practical application is still limited by challenges such as membrane fouling, relatively high energy requirements, complex fabrication, and the need for precise process control. In this context, more recent developments have focused on alternative electro-assisted approaches that aim to address some of these limitations while maintaining high removal and recovery performance [156, 157].

Electrocoagulation as an Electro-Assisted Phosphorus Removal Pathway

Electrocoagulation (EC) is an important electrochemical pathway for phosphorus removal and serves as a useful bridge between conventional chemical precipitation and more advanced electrically driven separation technologies. Unlike conventional coagulation, in which metal salts are externally added, EC generates coagulant species in situ through the electrochemical dissolution of sacrificial electrodes, most commonly Fe or Al. The released metal ions

undergo hydrolysis to form metal hydroxides and other reactive species that interact with phosphate through adsorption, complexation, co-precipitation, and precipitation. The resulting phosphorus-containing flocs can subsequently be separated by sedimentation, flotation, or filtration. The sequence of anodic metal dissolution, phosphate precipitation, floc formation, and subsequent separation is schematically illustrated in Fig. 7(b and c) [142].

More broadly, electrocoagulation involves several simultaneous physical, chemical, and electrochemical pathways. Mousazadeh et al. [158] identified charge neutralization, precipitation or co-precipitation, oxidation–reduction reactions, adsorption onto metal hydroxides, and flotation as important contaminant-removal mechanisms in EC/ECF systems. However, the relative contribution of each pathway depends on the electrode material, target contaminant, pH, and operating conditions. For phosphate specifically, adsorption onto freshly generated metal hydroxides and phosphate co-precipitation have frequently been proposed as important pathways, rather than being definitively established as universal mechanisms for all EC systems [158].

The performance of EC is strongly influenced by electrode material, current density, solution pH, conductivity, electrode spacing, treatment time, and initial phosphorus concentration. Fe- and Al-based electrodes are particularly attractive because the generated iron and aluminum species exhibit a strong affinity toward phosphate. A systematic assessment of EC/ECF studies identified current density (typically 1–20 mA cm⁻²) electrode material, and electrolysis time as major performance-controlling parameters [158]. Nevertheless, the optimum values vary considerably with wastewater composition and reactor configuration, meaning that a single operating range should not be interpreted as universally applicable. Although EC can achieve high phosphorus removal without continuous addition of chemical coagulants, its practical performance should not be assessed solely on the basis of removal efficiency. Sacrificial-electrode consumption, electrical energy demand, electrode passivation, sludge generation, and the characteristics of the resulting phosphorus-containing solids are important considerations for long-term operation and scale-up [142, 158, 159].

The sensitivity of EC performance to electrode selection was demonstrated by G. Li et al. [160], who compared Al, Fe, and Mg electrode systems under controlled operating conditions. Under one reported test condition, 20 min of treatment at a current density of 3 mA cm⁻², the authors observed phosphate removals of 90%, 80%, and 35% using Al, Fe, and Mg systems, respectively. However, the relative performance varied with electrode spacing, pH, current density, and treatment time, indicating that these values reflect

condition-specific outcomes rather than universal rankings of electrode materials.

Importantly, they also evaluated the environmental implications of electrode selection through life-cycle assessment. The Al system exhibited lower environmental impacts than the Fe and Mg systems for most evaluated midpoint and endpoint indicators, whereas Mg contributed more strongly to most impact categories, apart from freshwater ecotoxicity. The study therefore demonstrates that the preferred electrode cannot be identified from phosphate-removal percentage alone; material consumption, electricity demand, sludge generation, treated-water quality, and wider environmental impacts must also be considered [160].

From a resource-recovery perspective, EC also demonstrates the important distinction between phosphorus removal and phosphorus recovery. Although phosphate can be efficiently transferred from the aqueous phase into Fe- or Al-containing precipitates, this does not necessarily produce a directly reusable phosphorus product. The composition, purity, and subsequent processing requirements of the generated solids therefore determine whether EC functions primarily as a phosphorus-removal process or can contribute to a broader recovery pathway. In this respect, EC differs from electro-separation technologies such as electrodialysis, CDI, MCDI, and FCDI, in which electrically controlled ion transport or reversible electrosorption can facilitate phosphorus concentration and regeneration [161].

This distinction is consistent with the broader electrochemical framework described by Y. Ren et al. [27]. Electrocoagulation commonly transfers phosphorus into Al- or Fe-rich solids, whereas electro-induced precipitation may recover phosphorus in more readily reusable forms, such as struvite or calcium phosphate. Thus, even when two systems report similar phosphorus-removal efficiencies, their recovery value may differ substantially depending on the concentration, purity, chemical form, and subsequent usability of the phosphorus-containing product.

Hybridization can further extend the application of EC by combining in situ coagulation and precipitation with membrane-based polishing. For example, EC-MBR configurations have demonstrated very low residual phosphorus concentrations and stable long-term operation, although sludge management and reactor integration remain important practical considerations [162].

Beyond membrane-based configurations, EC has increasingly been combined with advanced oxidation processes, including electrooxidation, ozonation, H₂O₂-assisted treatment, sulfate-radical processes, electro-Fenton reactions, and UV irradiation. Hajalifard et al. [163] reported that these EC-AOP combinations can improve the treatment of recalcitrant contaminants by coupling coagulation and

solid–liquid separation with oxidative degradation. At the same time, their review emphasized that evidence from real wastewater, continuous operation, and industrial-scale systems remains limited. Therefore, improvements observed in controlled laboratory experiments should not automatically be interpreted as evidence of large-scale feasibility.

A phosphorus-specific example of this hybrid approach was reported by Fekadu et al. [164], who investigated a peroxi-photoelectrocoagulation process combining EC with UV irradiation and H_2O_2 . Under the tested conditions, phosphate removal ranged from 38.46% to 97.91%, with the final concentration reduced to 0.12 mg P/L as $\text{PO}_4\text{-P}$. Reported energy consumption ranged from 0.27 to 5.94 kWh/g P and varied with initial phosphate concentration, pH, reaction time, current density, and H_2O_2 dosage. These results demonstrate the potential of EC-based systems for phosphorus polishing, while also showing that high removal can be accompanied by substantial variation in energy demand [164].

Taken together, these findings indicate that EC performance should be evaluated using a combination of removal efficiency, electrode consumption, energy demand, sludge characteristics, environmental impact, and phosphorus product quality. Such an evaluation is particularly important for hybrid systems, where the improved removal achieved by adding oxidation or membrane-separation stages must be weighed against greater equipment complexity, energy requirements, and process-control demands.

Capacitive Deionization and Membrane Capacitive Deionization

In contrast to electrocoagulation, which primarily transfers phosphorus into a coagulated or precipitated solid phase, capacitive deionization (CDI) and membrane capacitive deionization (MCDI) rely on electrically controlled and reversible ion adsorption. MCDI further incorporates ion-exchange membranes into the electrode configuration to improve ion selectivity and charge efficiency. Among the various electro-assisted technologies, CDI and its advanced form, MCDI, have emerged as particularly promising approaches for selective phosphorus removal and recovery. The most basic operation of CDI is applying an electrical potential to a pair of electrodes, most commonly made of porous carbon materials, to electro-sorb ions from water. The working mechanism comprises reversible attraction of ions to oppositely charged electrodes upon application of voltage, thereby effectively removing the solution from the stream. The whole process is then made reversible, with the possibility of regenerating the electrodes and removing adsorbed ions, thereby positioning CDI as a framework-based technology for the sustainable treatment of water [141, 165].

However, in conventional CDI systems, co-ion expulsion and competition from background electrolytes can significantly reduce phosphate selectivity, thereby motivating the development of membrane-assisted configurations such as MCDI [131]. Recent advancements in flow-electrode capacitive deionization (FCDI) further extend this concept by enabling continuous operation and simultaneous phosphorus removal and recovery, with reported removal efficiencies of approximately 92%, addressing one of the key limitations of conventional batch CDI systems [166]. The fundamental ion-removal mechanisms and operational differences among CDI, MCDI, and FCDI systems are schematically illustrated in Fig. 7(d–f). In conventional CDI (Fig. 7(d)), ions are removed through electro-sorption onto oppositely charged porous carbon electrodes; however, co-ion expulsion and limited selectivity reduce overall efficiency. The introduction of ion-exchange membranes in MCDI (Fig. 7(e)) improves charge efficiency and ion selectivity by suppressing co-ion transport. In contrast, FCDI (Fig. 7(f)) employs flowable slurry electrodes, enabling continuous operation and simultaneous ion adsorption and regeneration, thereby overcoming the batch limitations of traditional CDI systems [141].

As an example of recent work on CDI in wastewater treatment, rocking-chair CDI has been developed to target selectivity improvement in the PO_4^{3-} removal and recovery process. In this setup, the ion-exchange membranes are used to reject chloride ions and selectively take up PO_4^{3-} ions. The experimental results proved that around 90% of chloride was removed, while PO_4^{3-} remained up to 80% of its initial concentration when the process conditions were optimized. The electrodes used were carbon-based and Prussian blue analogues (PBA), and the experimental tests with both electrode ions revealed that the PBA electrode was successful in chloride removal with a higher removal rate than the carbon-based one. Low energy consumption and electrode material versatility can be named as key advantages of PBA electrodes in ion recovery from wastewater by rocking-chair CDI [167].

Another improvement in the CDI process is MCDI, which utilizes ion-exchange membranes to enhance the selectivity of ions and the capacity for adsorption [165]. MCDI has been reported to possess great promise for removing P from wastewater. High phosphorus removal and selectivity have been reported for MCDI under optimized voltage, flowrate, and pH conditions; however, performance remains dependent on wastewater composition, competing ions, electrode characteristics, and the initial phosphate concentration. In addition, recovering P via electrode regeneration will further enhance the sustainability and economic feasibility of this treatment [165]. Compared with conventional membrane or adsorption systems, MCDI offers the distinct advantage

of electrically driven, reversible ion removal, enabling both selective separation and efficient regeneration. This feature positions MCDI as one of the most promising next-generation technologies for phosphorus removal and recovery [168].

Recent developments and progress in MCDI, a sub-category of CDI, have attracted considerable attention due to its unique properties in desalination and nutrient recovery applications. The experimental results for integrating an MCDI based on an ion-exchange membrane (IEM) confirmed that the combination significantly improves desalination performance, for example, efficiency, owing to increased co-ion repulsion and Faradaic reactions. This resulted in excellent salt removal and higher charge efficiency. As an example, the MCDI systems consumed significantly less energy to achieve a charge efficiency of up to 67.79% compared to a membrane-free CDI system, namely 0.99 kWh/kg against 2.89 kWh/kg [169].

One innovative approach that is becoming popular is the combination of a membrane bioreactor (MBR) and an MCDI system. It has typically been used for nutrient removal and recovery from aqueous solutions. This approach consists of aqueous solution (urine) stabilization, leveraging the MBR process to convert volatile ammonia into the non-volatile forms NH_4^+ and NO_3^- . The benefit of combining these two processes is to achieve a high reduction in total dissolved solids (TDS) for NO_3^- , PO_4^{3-} , and NH_4^+ , as well as better removal rates for species that cannot be directly removed and recovered in an MCDI process. The approach produced recovery rates of 80% and 64% for NO_3^- and PO_4^{3-} , respectively. It is worth mentioning that this method demands only 14.75 kWh/kg of energy for the recovery process, which can be considered a more sustainable system compared with other nutrient recovery methods [170].

In search of other innovative methods, a covalent organic framework-membrane coating electrode (COF-MCE) was used for a novel solid-state P adsorption and recovery. The method consists of depositing a mixed polyvinyl alcohol (PVA) and polyethyleneimine (PVE) membrane onto activated carbon fiber and the in-situ growth of COF-LZU1. The system exhibited high salt adsorption capacity, excellent performance, excellent selectivity, and desirable features for P removal and recovery. Regarding the experimental results, the COF-MCE electrodes can perform 50 cycles without a notable decline in performance and can efficiently recover P at a reversal voltage of -0.8 V. It is important to note that the system is still too complex to be prepared on a large scale. It has some drawbacks regarding membrane fouling, which needs more investigation into how to enhance its characteristics [56].

Integrating electrodialysis with a membrane contactor resulted in the development of a four-chamber electrochemical membrane (4 C-EMS) system, which is capable of removing and recovering N and P from digester centrate. The system achieved phosphorus recovery efficiencies above 85%. Its specific energy consumption, when normalized to phosphorus recovery, ranged from 178.0 to 278.8 kWh/kg P across the tested operating conditions. The process also reduced heavy-metal concentrations in the recovered struvite, thereby improving product quality [171].

To achieve higher efficiency and selectivity for PO_4^{3-} removal and recovery from wastewater, a ZnFe-PANI/CNT composite electrode was used in an MCDI system. The electrode material is an assembly of ZnFe-LDHs, pristine polyaniline, and carbon nanotubes. The experimental results showed that the system can remove up to 90% of PO_4^{3-} even in the presence of competing ions. The electrode exhibited sufficient activity for the removal and recovery processes across a wide pH range. It has also been shown to exhibit rapid adsorption and low energy consumption. It can be considered a cost-effective candidate for wastewater treatment processes. On the other hand, like the other systems, it suffers from a complex procedure for synthesis and might encounter some challenges in terms of large-scale application [172].

Moreover, recent research has used a membrane capacitive deionization device with a reservoir (R-MCDI) to circulate electrolytes in the tiny-volume electrode chambers. The system was found to have significantly higher P removal rates compared with a conventional MCDI. The P removal rate of $0.052 \mu\text{mol}/(\text{cm}^2 \cdot \text{min})$ was achieved using R-MCDI, approximately twice that of conventional MCDI. Off-flow desorption operation in R-MCDI allowed continuous P enrichment, thereby reducing energy usage and supporting sustainable P recovery [168].

The progressive development of CDI-based technologies reflects a transition from conventional non-Faradaic electro-sorption systems toward more advanced Faradaic and hybrid configurations with enhanced selectivity and charge efficiency. As illustrated in Fig. 8(a), this evolution includes CDI, MCDI, FCDI, and HCDI systems, each representing a step forward in addressing key challenges such as co-ion interference, energy consumption, and long-term operational stability. This progression highlights the growing importance of material design and system integration in improving phosphorus selectivity and recovery under complex wastewater conditions [15].

To provide a broader comparison of electro-driven phosphorus recovery technologies, Fig. 8(b) summarizes the relationship between phosphorus enrichment rate and energy consumption based on reported literature data. CDI

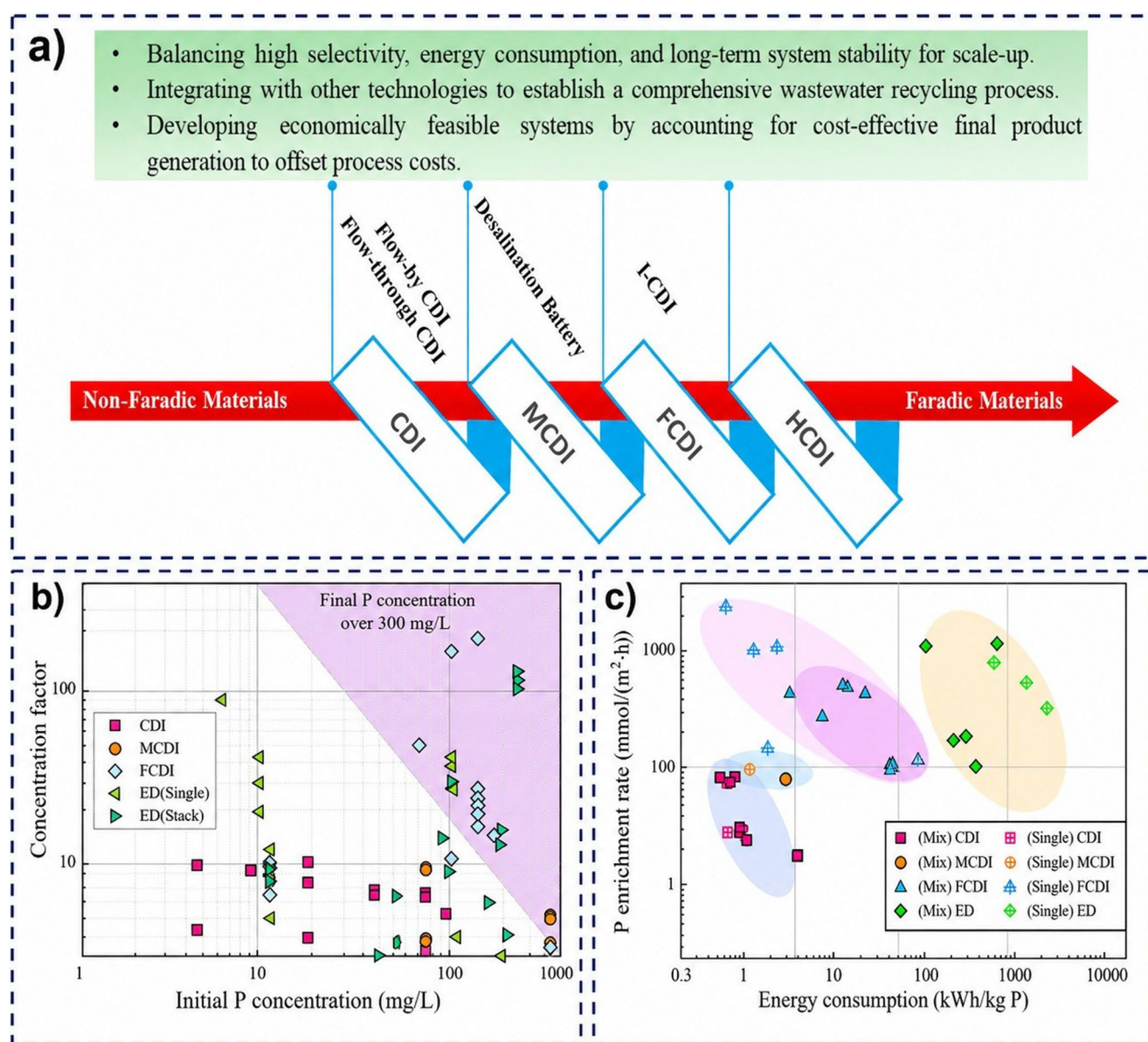


Fig. 8 (a) Evolution of CDI technologies from non-Faradaic to Faradaic systems and their associated challenges for nutrient (N and P) recovery in sustainable water treatment [15], (b) Enrichment rate vs.

energy consumption for CDI, MCDI, FCDI, and ED [34], and (c) Initial phosphorus concentration vs. concentration factor, indicating the recovery threshold (> 300 mg/L) [34]

and MCDI systems generally exhibit low energy consumption but relatively limited enrichment rates, primarily due to their intermittent operation and reliance on electro-sorption mechanisms. In contrast, electrodialysis (ED) achieves significantly higher enrichment rates, albeit at the expense of substantially increased energy demand. Between these extremes, FCDI demonstrates a more balanced performance, offering higher enrichment rates while maintaining moderate energy consumption. This comparison highlights the inherent trade-offs between efficiency and energy requirements, which remain a central consideration in the design and selection of electro-assisted phosphorus recovery systems [34].

Critical Insights and Future Directions for Membrane-based Phosphorus Recovery

A key limitation from a process engineering perspective is the ability of existing systems to achieve sufficiently high phosphorus concentrations for downstream recovery. As illustrated in Fig. 8(c), most CDI and MCDI systems operate at relatively low concentration factors, particularly at low initial phosphorus concentrations, making it difficult to reach the critical threshold (e.g., > 300 mg/L) required for efficient recovery processes. In contrast, electrodialysis-based systems and, to some extent, FCDI demonstrate a

greater capability to achieve higher concentration factors, although often at the expense of increased energy consumption and system complexity. This reflects a fundamental gap between laboratory-scale phosphorus removal performance and the practical requirements for resource recovery [34]. Accordingly, there is a clear need for technologies that not only remove phosphorus efficiently but also concentrate it to levels suitable for downstream recovery. In this context, recent integrated systems combining electrochemical, membrane, and biological processes have shown that coupling reaction and separation steps can enhance overall recovery efficiency while reducing system complexity [145, 150]. To provide a more consistent basis for comparison, Table 5 also reports the water matrix, initial and final phosphorus concentrations, pH, regeneration or recovery performance, and energy consumption, where available in the original studies. Because the relevant operating parameters differ among membrane, electrocoagulation, electrodialysis, and electrosorption systems, technology-specific information is reported where appropriate rather than imposing identical operating parameters across fundamentally different processes.

The chemical form of the recovered phosphorus also influences the practical value of electro-assisted systems. Y. Ren et al. [27] summarized an electrochemical precipitation column containing CaCO_3 particles that maintained phosphorus removal at approximately $90 \pm 5\%$ under optimized conditions, with reported energy consumption of 29–61 kWh/kg P. In this system, phosphorus was recovered as calcium phosphate, providing a clearer route to product reuse than transferring it into Fe- or Al-rich electrocoagulation sludge. A separate CaCO_3 -loaded carbon-nanotube filtration system achieved a reported phosphate recovery of 84.1% after 3 h at an applied voltage of 3 V [27]. Although these studies demonstrate the potential of electro-induced precipitation, their results remain dependent on reactor configuration, mass transfer, solution composition, and the availability of calcium or magnesium for mineral formation.

Energy demand also varies substantially with electrode material, phosphorus concentration, current density, and recovery pathway. Again Y. Ren et al. [27] summarized reported EC energy-consumption ranges of approximately 0.18–11.29 kWh/m³ for Al electrodes and 0.24–8.47 kWh/m³ for Fe electrodes. These broad ranges should not be treated as directly comparable performance benchmarks because they originate from systems operated under different conditions. Nevertheless, they illustrate why energy should be reported relative not only to the volume of water treated but also to the mass of phosphorus removed and, where possible, the mass of phosphorus actually recovered.

The expanded comparison also demonstrates that similar phosphorus-removal efficiencies can correspond to

substantially different process outcomes. In particular, the generation of a phosphorus-enriched concentrate, successful desorption or regeneration, or transfer of phosphorus into a precipitated solid should not be interpreted as equivalent recovery outcomes. Therefore, energy demand, regeneration performance, water matrix, and the final form and concentration of phosphorus need to be considered alongside removal efficiency when assessing the practical potential of these systems. As shown in Table 5 the performance of hybrid membrane systems is strongly governed by the dominant phosphorus-control mechanism: biologically driven and adsorption-based systems favor high removal efficiency but limited concentration capability, whereas electro-assisted and membrane-separation processes enable improved phosphorus concentration and recovery, albeit with increased energy demand and system complexity. This highlights that the selection of an appropriate hybrid system should be guided not only by removal efficiency, but also by the targeted recovery pathway and the desired form of phosphorus in the final product.

This distinction is particularly important when interpreting reported phosphorus-removal efficiencies. Membrane rejection, for example, may produce a phosphorus-enriched retentate without recovering phosphorus as a reusable product, while electrocoagulation may achieve high phosphate removal by transferring phosphorus into Fe- or Al-rich solids. Similarly, CDI/MCDI regeneration can release adsorbed phosphate into a concentrate, but this represents concentration or desorption rather than complete phosphorus recovery unless the phosphorus is subsequently isolated or converted into a reusable product. Consequently, the location, concentration, chemical form, and subsequent fate of phosphorus after treatment should be considered alongside removal efficiency when evaluating recovery-oriented technologies.

Despite these advances, a common challenge across electro-assisted membrane systems is balancing performance, selectivity, energy consumption, fabrication complexity, and operational robustness. Each of the methods discussed offers clear advantages, but these are often accompanied by practical limitations. In particular, improvements in removal efficiency do not necessarily translate into enhanced selectivity or scalability, underscoring the inherent trade-offs in current system designs.

Electro-assisted systems based on CDI and MCDI offer a promising pathway, as they enable reversible, electrically controlled ion removal. However, these systems remain highly sensitive to operational conditions such as pH, ionic strength, and the presence of competing ions, which can significantly influence adsorption behavior and charge efficiency in real wastewater applications. Furthermore, the presence of competing ions such as sulfate, nitrate,

Table 5 Comparison of membrane-based hybrid systems for phosphorus removal and recovery, including their dominant removal mechanisms, typical performance, key advantages, and practical limitations from a process engineering perspective

Hybrid membrane platform	Membrane role	Dominant <i>P</i> -control mechanism	Reported performance	<i>P</i> conditions and final performance	Recovery/re-generation & energy	Key advantages	Main limitations	Key insight	Ref.
A ² O/A+MBR	Solid-liquid separation and biomass retention	Biological phosphorus uptake by PAOs/DGAOs	P removal improvement	Municipal wastewater; influent TP 4.1 ± 1.1 mg P/L; pH NR; effluent TP 0.36 ± 0.32 mg P/L (advanced A ² O/A+MBR phase).	Direct P recovery/regeneration not demonstrated; phosphorus retained in biomass/sludge. Specific energy for P recovery: NR.	Mature and stable nutrient removal platform	Limited direct phosphorus recovery; sensitive to microbial and influent fluctuations	Effective for phosphorus removal, but not primarily a recovery-oriented platform	[143, 144]
AnMBR (Biom mineralization)	Biomass retention and anaerobic filtration	Microbial Fe-P biomineralization with vivianite formation	TP removal 96.8 ± 1.7%; methane production increased from 0.20 ± 0.05 to 0.29 ± 0.02 L CH ₄ ·g ⁻¹ COD	Swine wastewater; initial TP NR; pH NR; TP removal 96.8 ± 1.7% (Phase III) and 98.0 ± 0.9% (Phase IV).	P recovered in Fe-P biominerals, including vivianite; methane production increased to 0.29 ± 0.02 L CH ₄ ·g COD. Specific electrical energy: NR.	Simultaneous phosphorus removal and energy recovery	Process complexity; Fe balance and fouling control remain important	phosphorus recovery can be integrated with energy generation	[145]
Vermiculite-MBR	Membrane separation with mineral-assisted adsorption	Adsorption-enhanced biological removal	TP removal improvement from 35% to 74%	Synthetic wastewater; influent TP 87 mg/L; pH 6–8; TP removal increased from 35% to 74%.	Direct P recovery/regeneration not demonstrated; vermiculite regeneration and specific energy consumption: NR.	Low-cost material integration; practical improvement in MBR phosphorus removal	Limited direct recovery capability; regeneration not the main focus	Low-cost material choice can substantially improve removal without sophisticated electrochemical design	[147]
CMF (cathodic membrane filtration)	Reactive membrane surface	Electrochemically induced calcium phosphate precipitation	P removal efficiency up to 96.2%	Synthetic phosphate electrolyte; initial PO ₄ -P 0.6 mmol/L (~18.6 mg P/L); pH 7.0; P removal 96.2%.	P recovered as calcium-phosphate precipitate, with hydroxyapatite as the stable phase; energy consumption 45.7 kWh/kg P.	Combines filtration and in-situ precipitation	Scaling, fouling, and process control complexity	Membranes can function as reactive interfaces, not only as passive barriers	[144]

Table 5 (continued)

Hybrid membrane platform	Membrane role	Dominant <i>P</i> -control mechanism	Reported performance	<i>P</i> conditions and final performance	Recovery/re-generation & energy	Key advantages	Main limitations	Key insight	Ref.
EMR (electroactive membrane reactor)	Electroactive filtration medium	Phosphonate oxidation followed by Ca-P precipitation at the cathodic membrane interface	85% phosphonate degradation; 80% phosphorus recovery	Synthetic RO-concentrate matrix; initial ATPP 30 mg/L; pH 6; 85% phosphonate degradation and 80% P recovery.	80% P recovery as calcium-phosphate precipitate; energy consumption 19 kWh/kg P. Regeneration cycle not applicable.	Single-step conversion and recovery; low reported specific energy use	Long-term stability and broader validation in real concentrate matrices are still needed	Strong example of process intensification through simultaneous reaction and separation	[150]
ED-MB (bipolar membrane electrodialysis)	Ion-selective transport and separation	Electric-field-driven ion migration and nutrient concentration	NH ₄ ⁺ removal up to 96.5%; PO ₄ ³⁻ removal up to 75.4%	Simulated N/P wastewater; initial P 200 mg/L; pH NR; PO ₄ ³⁻ removal up to 75.4% after 40 min.	Phosphate was separated/enriched into the phosphoric acid chamber; isolated reusable P product not reported. Energy requirement reported as ~1.8 kWh/kg (study basis).	High selectivity and direct nutrient recovery potential	High energy demand and membrane cost	Useful for controlled nutrient separation, but the energy penalty remains important	[151]
EC-MBR hybrid	Membrane polishing with electrocoagulation support	Fe-based coagulation/precipitation plus biological synergy	Effluent TP < 0.02 mg/L; filtration efficiency 5.5–8.3 LMH; stable operation over 120 days	Simulated rural wastewater; influent TP ~ 1.5 mg/L; pH NR; final TP 0.011–0.013 mg/L (< 0.02 mg/L).	P transferred to insoluble Fe-phosphate precipitates; downstream P recovery/regeneration not quantified. System used solar-powered EC and gravity-driven membrane filtration; specific energy NR.	Strong polishing performance; good long-term stability	Sludge/precipitate management and reactor integration complexity	Coupling electrocoagulation with membrane-based treatment improves both phosphorus removal and operational robustness	[162]

Table 5 (continued)

Hybrid membrane platform	Membrane role	Dominant <i>P</i> -control mechanism	Reported performance	<i>P</i> conditions and final performance	Recovery/re-generation & energy	Key advantages	Main limitations	Key insight	Ref.
FO (MOF-interlayered membrane)	Selective osmotic membrane barrier	Size/charge rejection with improved interfacial transport	Water flux 32.13 L/m ² .h; 57% higher than pristine membrane; structural parameter reduced from 578.29 to 337.23 μ m; phosphorus rejection 99%	Synthetic phosphate feed; initial phosphate 100 ppm (0.143 g/L KH ₂ PO ₄); pH NR; phosphorus rejection 99.10–99.71% (~99%).	No isolated P recovery/regeneration demonstrated; P remained on the feed/retentate side while water permeated to the draw solution. Specific energy consumption: NR.	Very high rejection with improved permeability	FO still needs a complete recovery train; draw solution management and ICP remain relevant	Excellent membrane-engineering solution, but selectivity alone does not complete the recovery process	[140]
MCDI	Ion-exchange membrane-assisted electro-sorption	Electro-sorption with co-ion suppression	Charge efficiency up to 67.79%; energy consumption 0.99 kWh/kg versus 2.89 kWh/kg for membrane-free CDI in the cited comparison	P-specific operating conditions were not reported in the cited MCDI desalination study; the cited experiment used a synthetic Na ₂ SO ₄ electrolyte (200 mg/L in the base test).	Reverse-voltage desorption improved regeneration but increased energy demand; MCDI specific energy was 0.99 kWh/kg versus 2.89 kWh/kg for membrane-free CDI (salt-removal basis).	Reversible operation; improved selectivity relative to CDI	Limited concentration capability; still affected by competing ions	Promising balance of selectivity and energy efficiency, but still limited in achieving high final phosphorus concentrations	[141, 169]
FCDI	Membrane-guided flow-electrode system	Continuous electrosorption and ion transport	Phosphate removal ~92% with carbon black flow electrode; ~80% with activated carbon under the same conditions	Synthetic phosphate wastewater; initial P 100–300 mg/L and pH 5–8 in optimization tests; ~92% maximum phosphate removal with carbon black flow electrode (~80% with activated carbon).	Continuous electrosorption enables phosphate transfer/concentration in the flow electrode for recovery; a separate isolated-P recovery efficiency and specific energy value were not reported.	Continuous operation and better scale-up logic than batch CDI	Flow-electrode handling, pumping demand, and system complexity	Bridges the gap between batch electrosorption and continuous nutrient recovery	[166]

Table 5 (continued)

Hybrid membrane platform	Membrane role	Dominant <i>P</i> -control mechanism	Reported performance	<i>P</i> conditions and final performance	Recovery/re-generation & energy	Key advantages	Main limitations	Key insight	Ref.
MBR–MCDI hybrid	Sequential biological stabilization and electro-assisted recovery	MBR stabilization followed by electrosorption/desorption recovery	At 2 V adsorption: 66% nitrate, 49% phosphate, 58% ammonium; at –2 V desorption: 80% nitrate, 64% phosphate, 76% ammonium recovery	Source-separated urine after MBR; MCDI feed PO_4^{3-} 265 mg/L; pH 6.5; at 2 V, treated PO_4^{3-} 136 mg/L (~49% removal).	At –2 V, phosphate recovery was ~64% in concentrated brine; no chemicals required for adsorption/desorption. <i>P</i> -specific energy was NR; reported <i>N</i> -recovery energy was 3.03–11.25 kWh/kg $\text{NH}_4\text{-N}$ and 3.87–14.75 kWh/kg $\text{NO}_3\text{-N}$.	Clear division of treatment and recovery functions	More complex than a single-unit system; currently application-specific	Good example of task-sharing between biological pretreatment and electro-assisted nutrient recovery	[170, 173]
4 C-EMS (ED+membrane contactor)	Multi-chamber selective separation	Electro-driven nutrient recovery with improved product quality	<i>P</i> recovery > 85%; specific energy consumption: 178.0–278.8 kWh/kg <i>P</i>	Digester centrate containing 60–70 mg/L $\text{PO}_4^{3-}\text{-P}$ and ~800 mg/L $\text{NH}_4^+\text{-N}$; >85% <i>P</i> recovery and >95% <i>N</i> recovery	Acid generation in situ from electrolysis compensating for 10–32% of acid needs; specific energy consumption 178.0–278.8 kWh/kg <i>P</i> and 8.2 ± 0.2 kWh/kg <i>N</i>	Good recovery and improved nutrient product quality	Multi-chamber design and control burden increase complexity	Shows that product quality matters alongside recovery efficiency	[171]

NR: not reported in the original study. Reported phosphorus concentrations retain the phosphorus basis used in the original publications unless otherwise stated

and bicarbonate continues to limit phosphate selectivity in electro-assisted systems, underscoring the need to develop highly selective electrode materials and membranes that operate under realistic wastewater conditions. This sensitivity highlights the importance of evaluating system performance beyond controlled laboratory environments.

Although adsorption and electro-assisted technologies can all achieve phosphorus removal, the fate of the separated phosphorus differs substantially among these pathways. As summarized in Fig. 9, adsorption and electrosorption require subsequent regeneration to release captured phosphorus; electrocoagulation primarily transfers phosphorus into a metal-rich solid phase; and electrodialysis can directly enrich phosphorus in a concentrated liquid stream.

These differences strongly influence whether a process provides only phosphorus removal or can facilitate subsequent concentration and recovery.

Accordingly, high removal efficiency alone should not be interpreted as evidence of effective phosphorus recovery. Recovery-oriented evaluation should additionally consider the concentration and chemical form of the separated phosphorus, regeneration requirements, energy demand, stability over repeated operation, and the feasibility of producing a reusable phosphorus-containing stream or product.

Looking ahead, further progress will depend on integrating material design with process engineering rather than optimizing individual components in isolation. The development of selective electrode materials, fouling-resistant

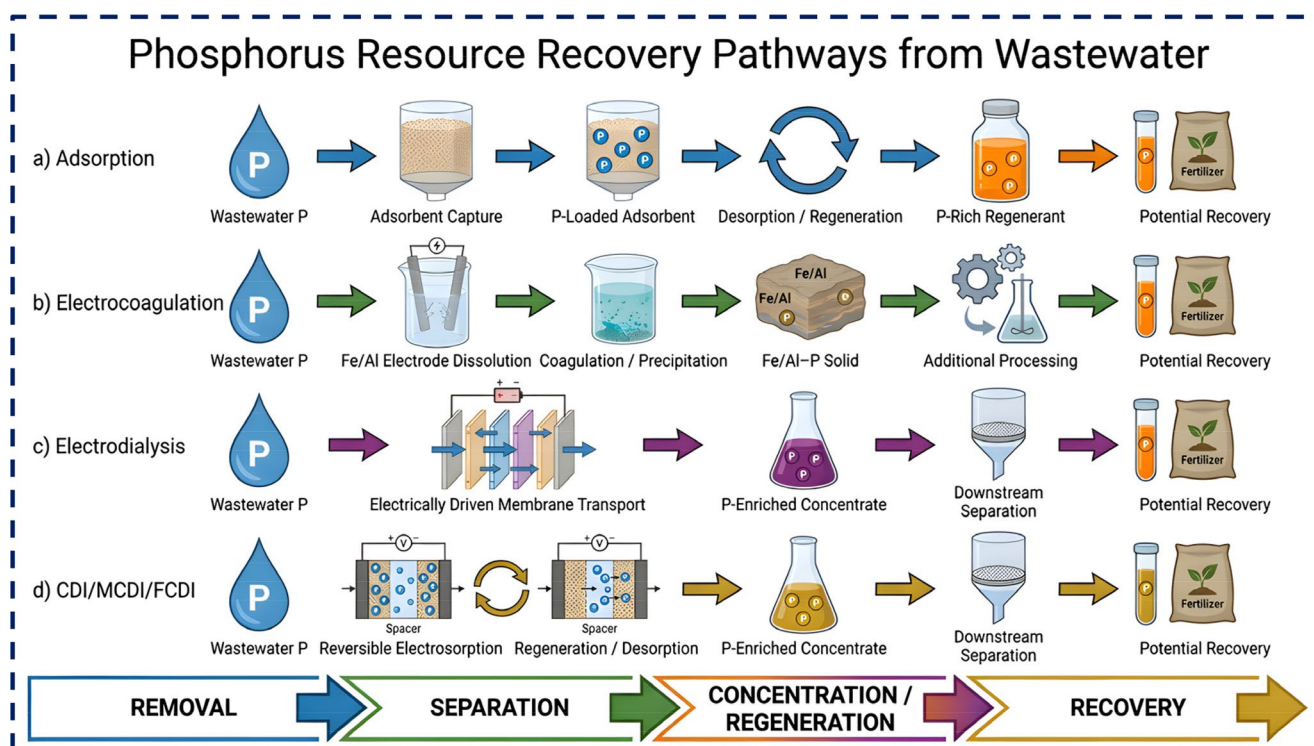


Fig. 9 Fate of phosphorus during adsorption and electro-assisted treatment: (a) adsorption, (b) electrocoagulation, (c) electrodialysis, and (d) CDI, MCDI, and FCDI

membranes, and well-integrated system configurations will be critical for improving overall performance. Approaches that combine adsorption, membrane separation, and electrochemical regeneration on a single platform are particularly promising, as they can simultaneously address multiple limitations and improve process efficiency. Future research should therefore focus on multifunctional system integration, in which adsorption, electrochemical transformation, and membrane separation are combined on a single platform to enable the simultaneous removal, concentration, and recovery of phosphorus in an energy-efficient manner.

The scale-up of hybrid electrochemical systems requires particular caution. Hajalifard et al. [163] found that most EC-AOP studies had been conducted at laboratory scale and that sufficiently comprehensive evidence from real wastewater and industrial operation was not yet available. Their review also identified only a small number of life-cycle assessments of EC-AOP combinations, despite the additional electricity, oxidants, electrodes, and reactor components required by these systems. In one reviewed comparison, sequential EC followed by ozonation showed a lower environmental burden than simultaneous EC-ozonation because the sequential arrangement shortened the required ozonation period [163]. Although this result supports thoughtful process sequencing, it remains study-specific and should not be interpreted as evidence that all sequential EC-AOP configurations are environmentally preferable.

Future studies should therefore compare stand-alone and hybrid systems using complete phosphorus mass balances, consistent energy metrics, electrode and chemical consumption, sludge or product quality, and life-cycle environmental impacts. Particular attention should also be given to whether the additional treatment stage improves only aqueous-phase removal or genuinely increases phosphorus concentration, regeneration efficiency, or product recovery.

Overall, electro-assisted membrane hybrid technologies represent a shift from passive separation processes to more controllable, integrated systems capable of simultaneous phosphorus removal and recovery. Nevertheless, their large-scale application remains limited by challenges such as membrane fouling, energy demand, fabrication complexity, and process control requirements. Bridging the gap between laboratory-scale performance and real-world application remains a key challenge, particularly under variable wastewater conditions where competing ions and fluctuating chemistry can significantly affect system performance.

Conclusion and Future Perspectives

This review highlights a continuing shift in phosphorus management from conventional removal toward integrated strategies that combine effluent polishing with phosphorus concentration and resource recovery. Chemical precipitation

and biological treatment remain important, established technologies because of their reliability and scalability, whereas adsorption, membrane-based separation, electrocoagulation, electrodialysis, and CDI-based systems offer distinct opportunities for selective phosphorus capture, concentration, and subsequent recovery. Importantly, however, high phosphorus removal should not be considered equivalent to phosphorus recovery. The practical value of these technologies depends on the form and concentration of the separated phosphorus, the feasibility of regeneration or downstream processing, and the ability to produce a reusable phosphorus-containing stream or product.

The comparison presented in this review also reveals that no single emerging technology simultaneously maximizes removal efficiency, selectivity, regenerability, energy efficiency, and recovery potential. Adsorbents can provide high phosphate affinity and effective polishing, but strong binding may compromise regeneration and long-term reuse. Electrocoagulation can achieve efficient phosphorus removal through in situ precipitation and coagulation, although electrode consumption, energy demand, sludge generation, and the subsequent usability of the phosphorus-containing solids remain important considerations. Electrodialysis can achieve higher phosphorus concentrations, but generally at increased energy and system complexity, whereas CDI, MCDI, and FCDI offer electrically controlled and potentially reversible ion capture but remain affected by competing ions, electrode stability, and limitations in achievable concentration. These trade-offs indicate that future development should focus on matching treatment mechanisms to the intended phosphorus endpoint rather than optimizing removal percentage alone.

Several research priorities are particularly important for translating these technologies toward practical application. First, future studies should move beyond short-term batch experiments toward continuous-flow operation under realistic hydraulic and phosphorus-loading conditions. Second, performance should be validated using real municipal, industrial, and agricultural wastewaters containing competing ions, dissolved organic matter, suspended solids, and variable phosphorus concentrations. Third, adsorbents, membranes, and electro-assisted systems should be evaluated over multiple regeneration and reuse cycles to quantify capacity retention, material degradation, fouling, electrode passivation, and changes in selectivity. Fourth, complete phosphorus mass balances should be reported to identify phosphorus remaining in the treated effluent, retained within the treatment material or solid phase, released during regeneration, concentrated in a recovery stream, and ultimately recovered as a usable product.

Energy and resource requirements also require more consistent assessment. For electro-assisted technologies in particular, energy consumption should be reported relative to the mass of phosphorus removed and, more importantly, the mass of phosphorus actually recovered, rather than only per volume of wastewater treated. Future studies should similarly report consumable requirements, including electrode consumption, regenerant use, membrane replacement, and chemical inputs, to enable meaningful techno-economic and environmental comparisons. Standardized reporting of initial and final phosphorus concentrations, phosphorus species, recovery efficiency, concentration factor, regeneration performance, and energy consumption would substantially improve comparison among emerging technologies.

Finally, pilot-scale studies are needed to determine whether the performance demonstrated under laboratory conditions can be maintained during long-term operation. Particular attention should be given to hybrid systems that strategically combine complementary functions, for example, adsorption for selective polishing, electrochemical processes for controllable separation or regeneration, and membranes for concentration and product-stream management. Such integration may help bridge the current gap between stringent phosphorus discharge requirements and circular nutrient management. The next stage of development should therefore be judged not only by how effectively phosphorus disappears from the treated water, but by how selectively, efficiently, and sustainably it can be concentrated and returned to use.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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